Ext

9

In this chapter, we will learn about

- uniform circular motion
- centripetal acceleration
- centripetal force and its daily examples

Uniform Circular Motion

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Ext

9.1 Describing circular motion

Flipped classroom

Watch a video and answer the questions.

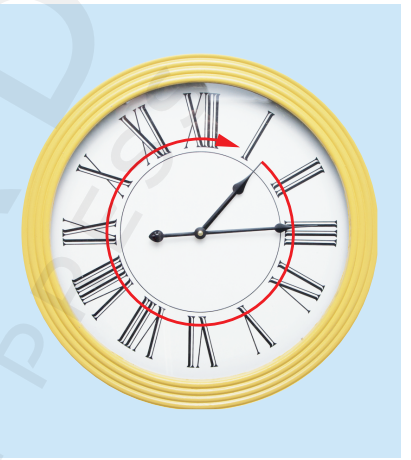
DSE goal

Define angular velocity as the rate of change of angular displacement and relate it to linear velocity.

You can easily find objects moving in circles in daily life (Fig 9.1a). We say that these objects perform circular motion.

(i)

(ii)



(iii)

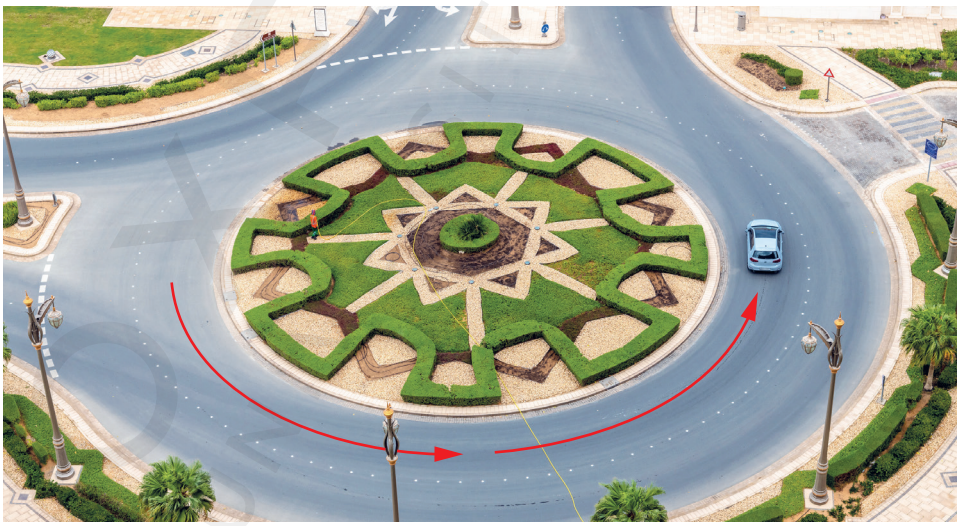


Fig 9.1a *Objects moving in circles (i) cabins of a Ferris wheel, (ii) tips of clock hands, (iii) cars travelling at a roundabout.*

Let us learn some quantities that are often used in describing circular motion.

1 Radian as a unit of angle

In circular motion, angles are measured in an SI unit called the **radian** (**rad**). An angle θ in radians is the ratio of the arc length s to the radius r (Fig 9.1b on p.371).

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}$$

radian (rad) 弧度

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Since the circumference of a circle of radius r is $2\pi r$ (Fig 9.1c), the angle around a point θ_{circle} can be expressed in radians as:

$$\theta_{circle} = \frac{2\pi r}{r} = 2\pi$$

The unit 'rad' is often omitted when the angle is expressed in terms of π.

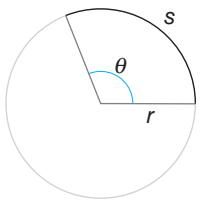


Fig 9.1b $\theta = \frac{s}{r}$.

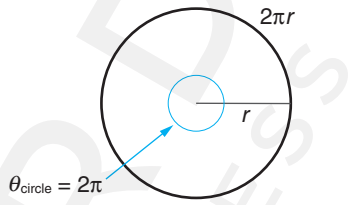


Fig 9.1c *Angle around a point = 2π.*

The radian is related to the degree (°) by

$$\frac{\theta \text{ (in radian)}}{2\pi} = \frac{\theta \text{ (in degree)}}{360°}$$

Table 9.1a shows the conversion between the two units of some angles.

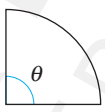
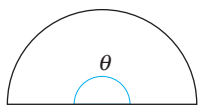
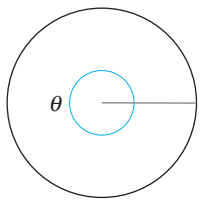
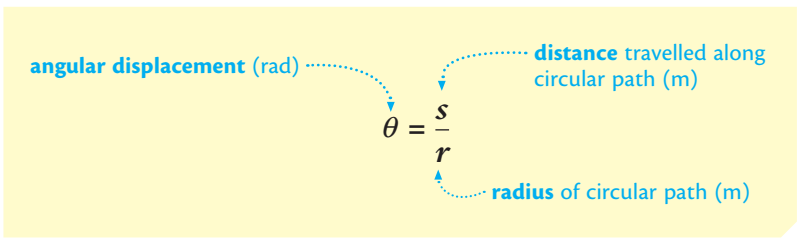
Angle			
Expressed in °	90°	180°	360°
Expressed in rad	$\frac{\pi}{4}$	π	2π

Table 9.1a *Conversion between radian and degree.*

The calculation for circular motion is much simpler when radian is used instead of degree.

2 Angular displacement

Consider an object moving from P to Q along a circular path of radius r (Fig 9.1d). The position of the object rotates through an angle θ which is the object's **angular displacement**. The magnitude of θ is written as:



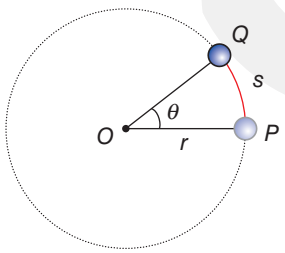


Fig 9.1d *An object has an angular displacement of θ.*

angular displacement 角位移

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The angular displacement concerns only the angle but not the distance travelled. For example, in Figure 9.1e, A and B move different distances d but they have the same angular displacement θ. An object has an angular displacement of 2π when it completes one revolution (Fig 9.1f).

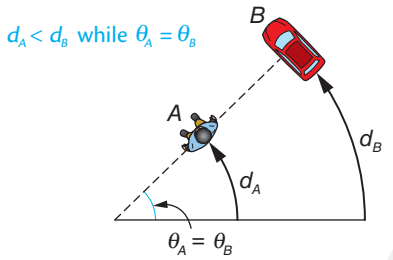


Fig 9.1e *Objects moving different distances can have the same angular displacement.*

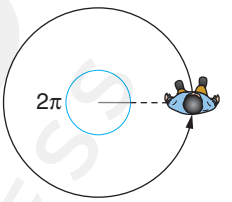


Fig 9.1f *The angular displacement for a revolution is 2π.*

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Complete Table a to convert angles between degrees and radians.

Angle (in degree)	45°		165°		321°
Angle (in radian)		$\frac{\pi}{6}$		2.62	

Table a

2 Complete Table b to convert angular displacement between radians and numbers of revolution.

Angular displacement (in radians)		$\frac{\pi}{2}$	
Angular displacement (in numbers of revolution)	1		0.4

Table b

3 Arrange angles a, b and c in ascending order.

$a = 280°$, $b = 3.14$ rad, $c = \frac{3\pi}{2}$

4 Objects X, Y and Z align on a straight line while they perform circular motion about point O. Figure a shows the paths they have travelled after a certain time. Determine whether each of the following statements is correct.

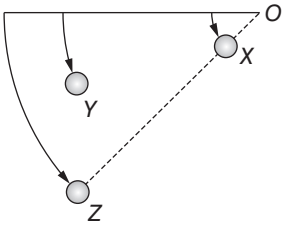


Fig a

(a) X and Z have the same angular displacement.

(b) Z has a larger angular displacement than Y.

(c) X and Z move the same distance.

(d) X moves a larger distance than Y.

Ext

3 Angular velocity and angular speed

We use **angular velocity** ω to describe the rate of change of angular displacement. It is defined as **the angular displacement per unit time**, i.e.

angular velocity (rad s^{-1}) ····· **angular displacement** (rad)

$$\omega = \frac{\theta}{t}$$

····· **time** of travel (s)

The magnitude of angular velocity can also be expressed in revolutions per second (rev s^{-1}):

$$1 \text{ rev } s^{-1} = 2\pi \text{ } s^{-1}$$

When an object performs **circular motion with a constant angular velocity**, such motion is called **uniform circular motion**.

Angular displacement and angular velocity are vectors. However, in this book, we are mostly concerned with their magnitudes and omit their directions for simplicity. The magnitude of the angular velocity is the **angular speed**.

Know more

Revolution per minute

Angular speed can be measured in the 'revolution per minute' (rpm). An object moving at 1 rpm completes one revolution per minute. In fact, this unit is commonly used in daily life. For example, the drum of a washing machine spins at 1200 rpm (Fig a), the fastest spin on ice skates is 342 rpm, etc.

The conversion between rpm and rad s^{-1} is given below:

$$1 \text{ rpm} = 2\pi \text{ min}^{-1} = \frac{2\pi}{60} \text{ s}^{-1}$$

Fig a *Spin speed in rpm shown on a washing machine.*

angular velocity 角速度　　uniform circular motion 勻速圓周運動　　angular speed 角速率

Ext

4 Linear speed

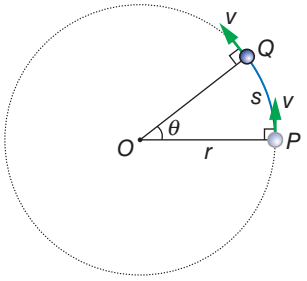


Fig 9.1g *An object performing uniform circular motion at a linear speed v.*

Consider an object performing uniform circular motion. At any point on its path, its speed v is called the ***linear speed***. It is equal to the magnitude of the object's linear velocity. The linear velocity at a point lies along the tangent of the circle at that point. Therefore, it keeps changing as the object moves (Fig 9.1g).

Suppose the object travels an arc length s from P to Q, and the radius of the circular path is r. Then the angular displacement θ of the object is equal to $\frac{s}{r}$ (Fig 9.1g). Rearranging the terms, we have:

$$s = r\theta$$

The linear speed v of the object is given by:

$$v = \frac{s}{t} = \frac{r\theta}{t} = r\left(\frac{\theta}{t}\right) = r\omega$$

linear speed (m s^{-1}) ····· **radius** of circular path (m)

$v = r\omega$

····· **angular speed** (rad s^{-1})

From the equation, we can note that:

- For objects moving with the same angular speed ω, the linear speed v is directly proportional to the radius of the path r (Fig 9.1h).
- If two objects have the same v, the one with smaller r has a greater ω (Fig 9.1i).

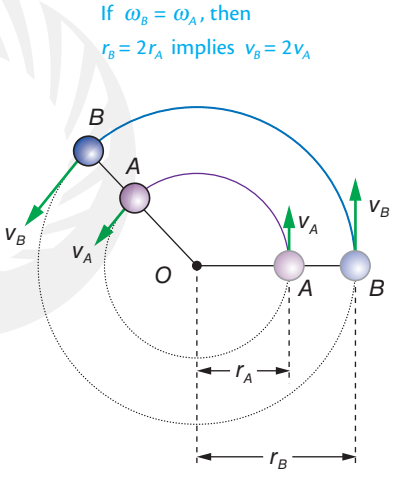


Fig 9.1h *If ω is fixed, v is directly proportional to r.*

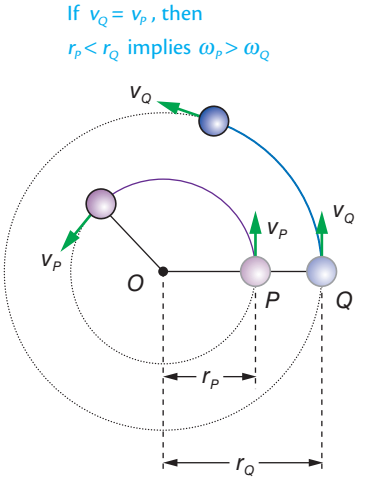


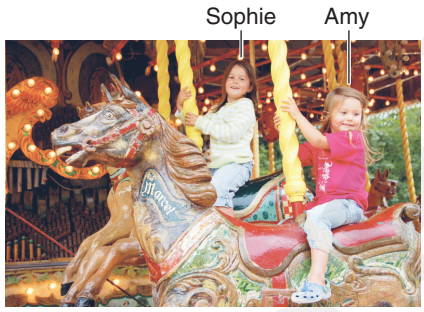
Fig 9.1i *P has a greater ω than Q if they move at the same v.*

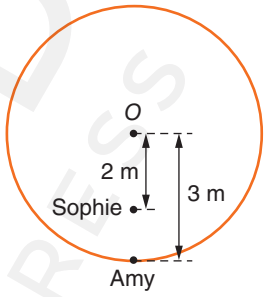
linear speed 線速率

Ext

Example 1 Angular speed and linear speed

Amy and Sophie are respectively 3 m and 2 m away from the centre O of a merry-go-round. They complete half a revolution in 10 s at a constant angular speed.





(a) Find their angular speed in rad s^{-1}.

(b) Find their angular displacement in 8 s. Express your answer in radians and degrees.

(c) Find their linear speeds.

Solution

Tip
θ for half a revolution = π

(a) Angular speed $\omega = \dfrac{\theta}{t} = \dfrac{\pi}{10} = 0.314$ rad s^{-1}

(b) $\theta = \omega t = \dfrac{\pi}{10} \times 8 = 2.51$ rad

Also,

$$\frac{\theta}{360°} = \frac{2.51}{2\pi}$$

$$\theta = 144°$$

(c) Amy's linear speed = $r\omega = 3 \times 0.314 = 0.942$ m s^{-1}

Sophie's linear speed = $r\omega = 2 \times 0.314 = 0.628$ m s^{-1}

5 Period

The **period** T of a uniform circular motion is **the time that the object takes to complete one revolution**. If the angular speed is ω, by $t = \dfrac{\theta}{\omega}$,

$$T = \frac{2\pi}{\omega}$$

period 週期

Ext

The distance that the object travels in time T is the circumference of the circular path. If the linear speed is v, by $t = \frac{s}{v}$,

$$T = \frac{2\pi r}{v}$$

Frequency f =
It is less commonly used in circular motion.

period (s) ····· $T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}$ ····· radius of circular path (m)

angular speed (rad s^{-1}) ····· linear speed (m s^{-1})

Quick check 2

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Complete the table by applying $v = r\omega$.

	r / m	ω / rad s^{-1}	v / m s^{-1}
(a)	0.23	1600π	
(b)		10.4π	65.3
(c)	5		12.5

2 *True or False:* An object performing uniform circular motion moves at a constant linear velocity. (T/F)

3 Objects X, Y and Z, initially on line L, start to perform uniform circular motion along different circular paths at the same time. Figure a shows their positions after a certain time interval. If $r_1 : r_2 : r_3$ is 1 : 2 : 4, determine whether each of the statements (a)–(c) is correct.

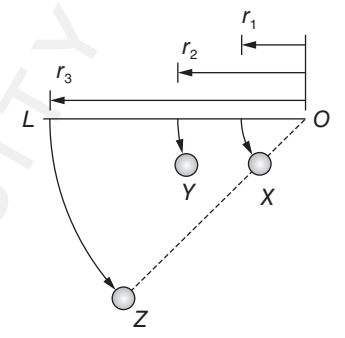


Fig a

(a) The linear speeds of X and Z are in a ratio of 1 : 4.

(b) The angular speeds of X and Y are in a ratio of 1 : 2.

(c) The periods of Y and Z are in a ratio of 1 : 2.

Learn by practice 1

1 An object performs a uniform circular motion of radius 2 m. Find the period if

(a) its angular speed is 3π s^{-1},

(b) its linear speed is 2.4 m s^{-1}.

2 An object performs a uniform circular motion with a radius of 0.2 m and a period of 30 s. Find its angular speed and linear speed.

3 Mark sits on a merry-go-round which rotates constantly at 0.5 rad s^{-1}. How long does it take for him to complete 3 revolutions?

4 A merry-go-round rotates uniformly. George is sitting on it at 5 m from its centre. The ride completes one revolution every 20 s.

(a) Find George's angular speed.

(b) Find George's linear speed.

Ext

Exercise 9.1

★**1** During a foot drill, a troop of soldiers need to march with perfect alignment to keep their shape. *P*, *Q* and *R* are three soldiers in the same row. They have to keep in line even when making a turn (Fig a).

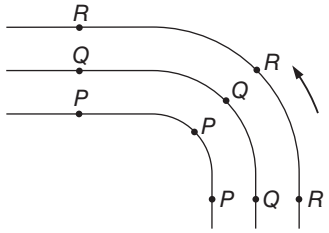


Fig a

Compared to *P* and *Q* when making the turn, *R* has the highest

(1) linear speed.

(2) angular speed.

(3) angular displacement.

A (1) only

B (2) only

C (1) and (2) only

D (2) and (3) only

2 An object travels a distance of 4 m in 2 s on a circular path of radius 5 m at a constant speed. Find the angular speed of the object.

3 Consider the second hand *S*, minute hand *M* and hour hand *H* of a clock (Fig b).

Fig b

Find the period *T* and the angular speed ω of each hand. Express ω in terms of π.

★**4** A car travels at a constant speed of 10 m s^{-1} around a roundabout of radius 20 m (Fig c).

Fig c

(a) Find the angular speed of the car.

(b) Find the angular displacement (in both radians and degrees) of the car in 3 s.

(c) How long does it take for the car to move around the roundabout and complete a revolution?

★**5** A cabin on the Ferris wheel moves at a uniform speed from *X* to *Y* as shown in 3 min (Fig d). The diameter of the wheel is 58 m.

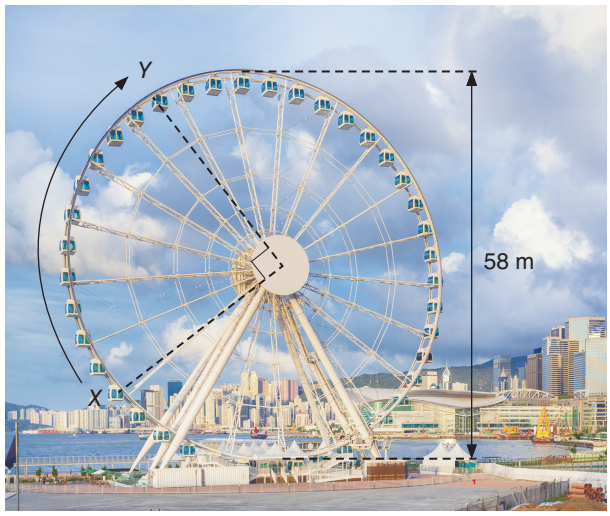


Fig d

(a) Find the period of the cabin.

(b) Find the angular displacement of the cabin in 10 min.

(c) Find the angular and linear speeds of the cabin.

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9.2 Centripetal force and centripetal acceleration

1 Centripetal force

DSE goal
Realize the resultant force pointing towards the centre of uniform circular motion.

In Figure 9.2a, a father is holding his son's hands and spinning him. It seems to him that his son is heavier than usual. Why is it the case?

Fig 9.2a *It seems to the father that the son is heavier while spinning.*

In the above case, the dad (object *X*) exerts a force on the son (object *Y*). This force together with the son's weight results in a net force which acts horizontally on *Y* (Fig 9.2b).

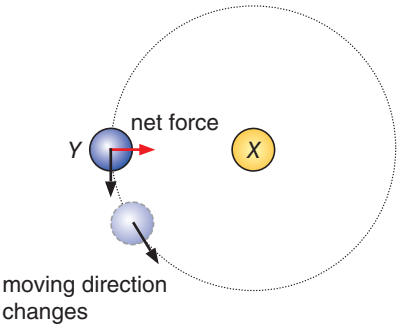

Fig 9.2b *Top view showing that the net force changes Y's moving direction.*

The net force is perpendicular to *Y*'s velocity, so no work is done on *Y*. *Y* does not speed up or slow down, but just changes its direction of motion. If this net force keeps acting on *Y*, *Y* will move in a circular path with a constant linear speed, i.e. in uniform circular motion. Note that this net force always points towards the centre of the circular path, so it is called the **centripetal force**.

The word 'centripetal' is originated from Latin, which means 'seeking the centre'.

centripetal force 向心力

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DSE goal

State centripetal acceleration $a = \frac{v^2}{r}$ and apply it to solve problem involving uniform circular motion.

DSE exam

16(1A)Q13, 20(1A)Q9, 20(1B)Q4(b)

2 Centripetal acceleration

According to Newton's second law, a net force acting on an object produces an acceleration in the same direction as the force. As the centripetal force acting on an object in uniform circular motion points towards the centre of the path, the acceleration a due to this force also points towards the centre and is called the **centripetal acceleration**. The magnitude of centripetal acceleration for uniform circular motion is given by (Fig 9.2c):

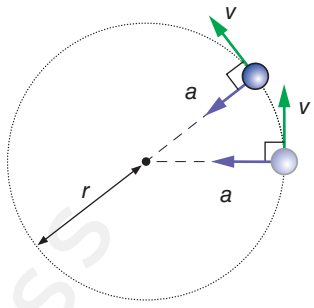


Fig 9.2c *Centripetal acceleration of an object in uniform circular motion.*

centripetal acceleration (m s^{-2}) — **linear speed** (m s^{-1})

$$a = \frac{v^2}{r} = r\omega^2$$

radius of circular path (m) — **angular speed** (rad s^{-1})

Know more

Concrete proof of centripetal acceleration for uniform circular motion

In chapter 1, we learned that acceleration is the rate of change of velocity.

$$\vec{a} = \frac{\Delta\vec{v}}{t} = \frac{\vec{v} - \vec{u}}{t}$$

When an object performs a uniform circular motion, the magnitude of its velocity remains constant while the direction of its velocity is changing.

Consider the movement of the object shown in Figure a.

During the movement, the object's velocity changes from $\vec{u}$ to $\vec{v}$. We can use the 'tip-to-tail' method to find $\Delta\vec{v}$ (Fig b). The triangle formed is isosceles. If the time interval for the movement tends to zero, $\Delta\theta$ also tends to zero and $\Delta\vec{v}$ is perpendicular to $\vec{u}$ (Fig c). Therefore, $\vec{a}$ is perpendicular to $\vec{u}$, i.e. the velocity of the object.

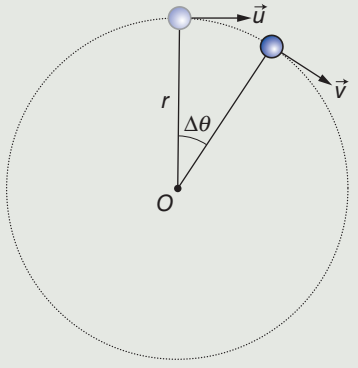


Fig a

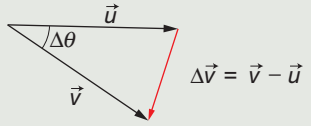


Fig b

Fig c

In Figure c, as $\Delta\theta$ is very small, Δv is approximately equal to the length of the arc.

$$\Delta v = v \times \Delta\theta$$

Divide both sides by time:

$$\frac{\Delta v}{t} = v \times \frac{\Delta\theta}{t}$$

$$a = v \times \omega$$

By substitute $v = r\omega$ or $\omega = \frac{v}{r}$, we have:

$$a = r\omega \times \omega = r\omega^2$$

$$a = v \times \frac{v}{r} = \frac{v^2}{r}$$

centripetal acceleration 向心加速度

Ext

→ This video demonstrates the Phone experiment.

Video

You may search the app on the Internet.

physics phone experiment app

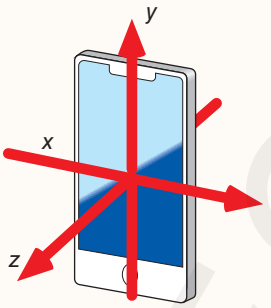
Let us do an experiment to verify the equation qualitatively.

Phone experiment

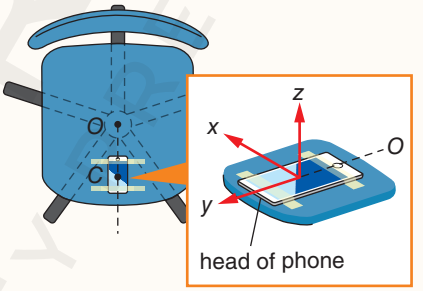
Centripetal acceleration and angular speed

What to do

1 Open an app that allows your phone to measure its acceleration along the *x*-, *y*- and *z*-directions and its angular speed during circular motion.



2 Fix the smartphone on a swivel chair with adhesive tape as shown. The *y*-axis of the phone should point radially away from the point *O*.



3 Start measuring the acceleration along the three directions. Spin the chair with the position of *O* fixed. Stop measuring when the spinning stops. Read the acceleration in *x*- and *y*-direction from the app.

4 Repeat step 3 by measuring the centripetal acceleration a and the angular speed ω. See the a–ω^2 graph obtained.

What you find

1 How does the acceleration in (a) x-direction and (b) y-direction change during the spin?

2 What is the relationship between a and ω^2?

The experiment results show that the acceleration always points towards the centre during the spin. It also shows that this acceleration decreases as the spinning of the chair slows down. This agrees with the equation $a = r\omega^2$.

The a–ω^2 graph obtained is a straight line passing through the origin. This shows that a is directly proportional to ω^2, which again agrees with the equation $a = r\omega^2$.

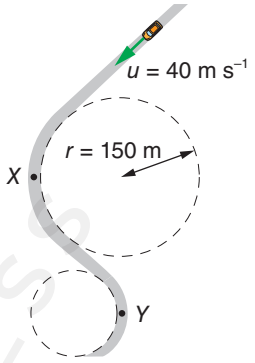
Ext

Example 2 | Centripetal acceleration and radius

A sports car races at a constant speed of 40 m s^{-1} on a track as shown. The track has two circular turns of different radii.

(a) Find the centripetal acceleration of the car at X.

(b) At which point, X or Y, does the car have a greater acceleration? Explain.



Solution

(a) Centripetal acceleration $= \frac{v^2}{r} = \frac{40^2}{150} = 10.7$ m s^{-2}

(b) The turn at Y has a smaller radius r. By $a = \frac{v^2}{r}$, the car has a greater acceleration a at Y for the same linear speed v.

Think about...

Simon argues that the car has a smaller acceleration at Y according to $a = r\omega^2$, since the radius at Y is smaller. Explain why his argument is wrong.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 An object performs uniform circular motion. Which direction does its acceleration point to?

A Same as its moving direction

B Towards the centre of its circular path

C Away from the centre of its circular path

2 (a) Andy and Betty jog along different circular paths with the same linear speed (Fig a). Who has a higher acceleration?

(b) Andy and Betty sit in a car which travels along a circular turn (Fig b). They have the same angular speed. Who has a higher acceleration?

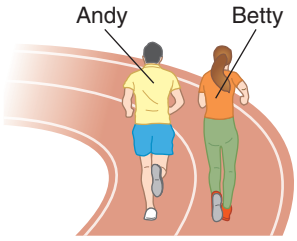


Fig a

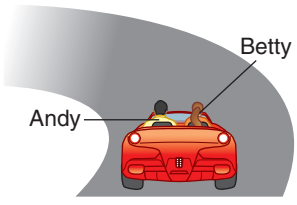


Fig b

Ext

Learn by practice 2

1 Find the centripetal acceleration of the object in each case.

(a) Object X moving in a circular path of radius 2 m at a constant speed of 0.25 m s^{-1}

(b) Object Y moving in a circular path of radius 12.5 m at a constant angular speed of 0.8π s^{-1}

2 An object performs uniform circular motion. Its acceleration is 2 m s^{-2} and the radius of its path is 1.2 m. Find its angular speed and linear speed.

☆ HKDSE 2016 Paper 1A Q13

→ This video demonstrates how this question is solved.

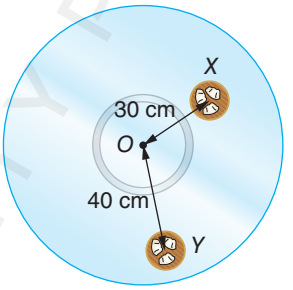
Video

DSE practice 1 | Acceleration at different positions

Dish X is 30 cm away from the centre O of a lazy Susan. Dish Y is 40 cm away from O. When the lazy Susan is turned at a constant angular speed, X and Y move along circular paths. What is the ratio of the acceleration of X to that of Y?

A 9 : 16 B 3 : 4

C 4 : 3 D 16 : 9



Solution

$a = r\omega^2 \Rightarrow a \propto r$ (for constant ω)

$$\frac{a_X}{a_Y} = \frac{r_X}{r_Y} = \frac{30}{40} = \frac{3}{4}$$

$\therefore$ The answer is B.

Common mistake ⚠

X and Y have the same ω but different v. Students may wrongly use the equation $a = \frac{v^2}{r}$ and get $a \propto \frac{1}{r}$.

Revision 9 Q9 (p.406)

DSE exam

19(1A)Q10, 21(1A)Q9, 22(1A)Q9

3 Circular motion and Newton's laws

a Circular motion and Newton's second law

By Newton's second law ($F = ma$), the magnitude of the centripetal force is

$$F = \frac{mv^2}{r} = mr\omega^2$$

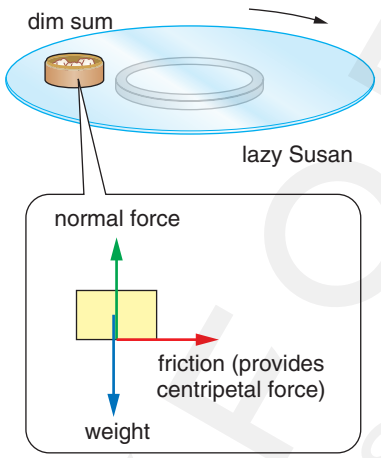
- m: mass (kg)
- v: linear speed (m s^{-1})
- F: centripetal force (N)
- r: radius of circular path (m)
- ω: angular speed (rad s^{-1})

Ext

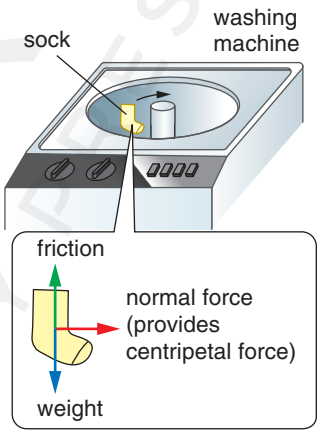
Note the following about the centripetal force for an object in uniform circular motion:

1 The centripetal force is the net force acting on an object to keep it in circular motion. The centripetal force can be provided by friction, gravitational force, etc. Figure 9.2d shows the origins of the centripetal forces in some situations.

(i) Dim sum rotating with a lazy Susan.



(ii) A sock rotated by a washing machine.



(iii) The Moon orbiting the Earth.

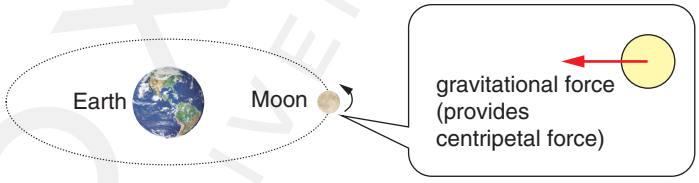


Fig 9.2d *Centripetal forces provided by different kinds of forces.*

2 The centripetal force and the centripetal acceleration are constant in magnitude, but their directions are constantly changing.

3 At any instant, the centripetal force is always perpendicular to the velocity of the object (Fig 9.2e). Therefore, the centripetal force **does no work** on the object.

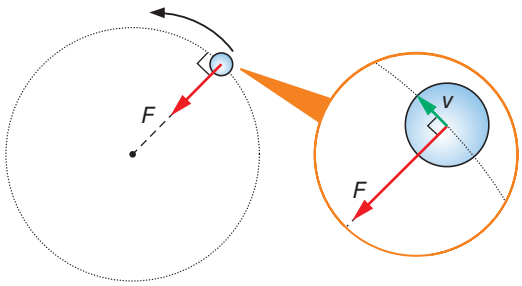


Fig 9.2e *The centripetal force is perpendicular to the velocity of the object.*

Ext

b Circular motion and Newton's first law

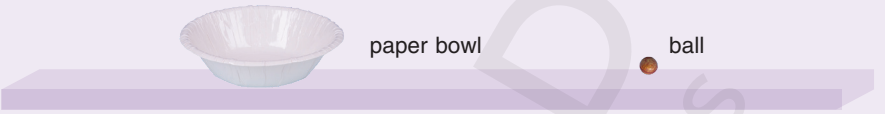
Video

→ This video demonstrates the Hands-on activity.

Leaving the circular track

Hands-on activity

What you need



What to do

1

Cut a section out of the bowl.

2

Push the ball gently along the wall of the bowl.

What you find

How does the ball move after leaving the bowl?

In the activity, the ball moves against the wall of the bowl and performs circular motion. The centripetal force is provided by the horizontal component of the normal force acting on the ball by the wall.

The normal force disappears once the ball leaves the bowl. According to Newton's first law, due to inertia, the ball continues to move at the same speed and direction as just before leaving the bowl. Its path becomes a straight line along the tangential direction (Fig 9.2f).

Video

→ This video shows the path of a ball leaving a circular rail.

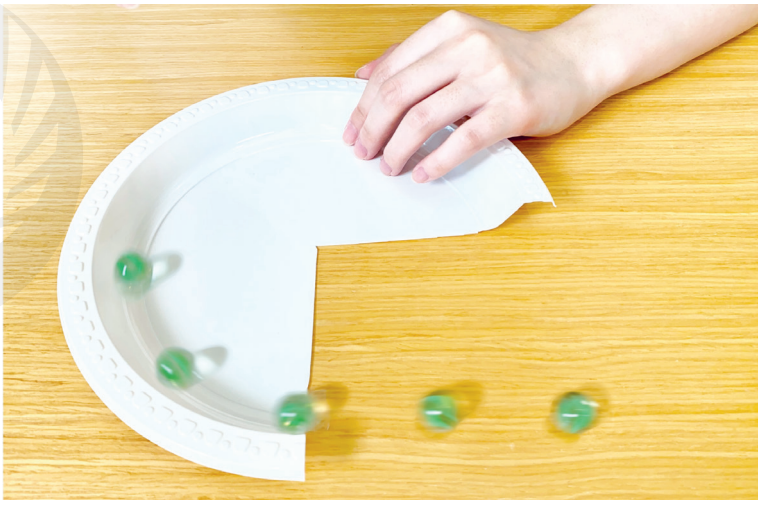


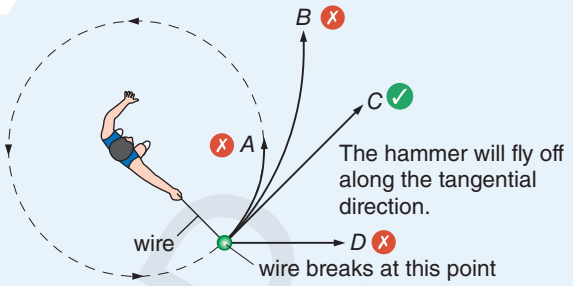
Fig 9.2f *The ball travels along a straight path after leaving the bowl.*

Ext

Right and wrong

An athlete swings the hammer in a circular path. The wire suddenly breaks during the swing.

Along which path will the hammer fly off?



Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 Complete the table below by applying $F = \frac{mv^2}{r}$.

	F	m	r	v
(a)		12 kg	2.5 m	0.4 m s^{-1}
(b)	2050 N		40 m	8 m s^{-1}
(c)	54 000 N	2500 kg		36 m s^{-1}
(d)	4 N	50 g	0.8 m	

2 Complete the table below by applying $F = mr\omega^2$.

	F	m	r	ω
(a)		6 kg	3.6 m	1.2 rad s^{-1}
(b)	84 N		20 cm	π s^{-1}
(c)	2820 N	7 kg		18.3 rad s^{-1}
(d)	12 500 N	250 kg	20 m	

3 An object performs uniform circular motion. Which of the following statements about is correct?

A The centripetal force should be drawn in the free-body diagram.

B The centripetal force equals to the resultant of all the forces in the free-body diagram.

C If there is more than one force in the free-body diagram, the centripetal force only equals to one of the forces.

4 A ball enters a frictionless horizontal circular channel from P (Fig a). Which arrow (A, B, C or D) best represents the path of the ball when it exits the channel at Q?

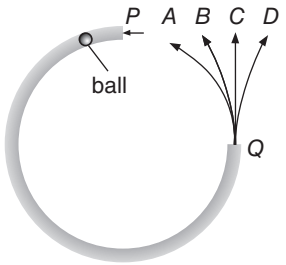


Fig a

Ext

Know more

Motion inside a turning car

When you sit in a car that turns left, you feel as if you are acted on by a so-called *centrifugal force* that pushes you towards the right. Does this force really exist?

Consider a bag placed on the car seat which is fairly slippery. The friction acting on the bag is nearly negligible. By Newton's first law, the bag keeps moving along a straight line when the car turns (Fig a). That explains why the bag slides towards the right when viewed inside the car. No centrifugal force acts on the bag!

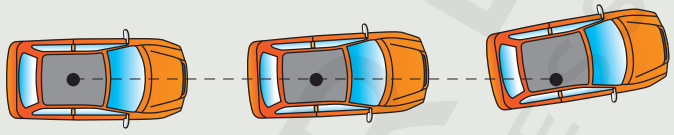


Fig a

Exercise 9.2

1 An object performs uniform circular motion. Which of the following statements about the acceleration of this object is/are correct?

(1) It points towards the centre of the circular path.

(2) Its magnitude is constant.

(3) If the object moves at a higher speed along the same path, the magnitude of the acceleration is larger.

A (1) only

B (2) only

C (1) and (3) only

D (1), (2) and (3)

***2** When an object moves at speed v along a circular path of radius r, the centripetal force it needs is F. If it moves at speed $2v$ along a circular path of radius $2r$, what is the centripetal force it needs?

A $\frac{F}{2}$

B F

C $2F$

D $4F$

3 An object moves in a circular path of radius 2 m at a constant speed of 1.2 m s^{-1}. If its mass is 5 kg, find the centripetal force for it.

4 An object moves in a circular path of radius 4 m at a constant angular speed of 0.5 rad s^{-1}. If its mass is 2.4 kg, find the centripetal force for it.

5 A ball moves on a horizontal floor at a constant speed along a circular path with centre O. The positions of the ball at different instants during the motion are shown (Fig a). In the figure, draw arrows to indicate the directions of the net forces acting on the ball at different instants.

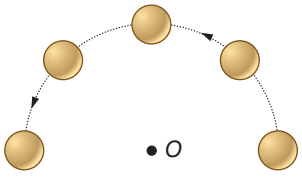


Fig a

***6** Figure b shows a 0.2-kg ball connected to a fixed point O by a 0.15-m long string on a smooth horizontal plane.

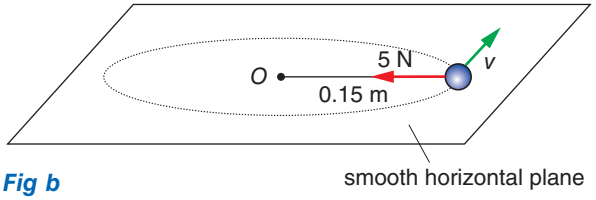


Fig b

The ball performs uniform circular motion at a constant speed v. The tension in the string is 5 N.

(a) Draw a free-body diagram for the ball.

(b) Which force provides the centripetal force?

(c) Find v.

centrifugal force 離心力

Ext

9.3 Examples of uniform circular motion

DSE exam

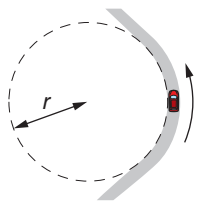
12(1B)Q3, 15(1A)Q10, 17(1B)Q5, 19(1B)Q4(a), 22(1B)Q3(c), 23(1B)Q4

1 Making turns on level roads

When a car of mass m turns a corner at a constant speed v along a circular path (Fig 9.3a), a centripetal force is required. On a level road, the centripetal force comes from the sideways friction f, which acts laterally between the road and the car (Fig 9.3b). If the radius of curvature of the turn is r,

$$f = \frac{mv^2}{r}$$

The equation can be written as $f = mr\omega^2$ if ω is given.



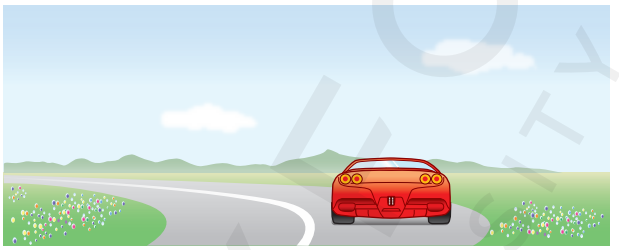


Fig 9.3a A car performs circular motion on a turn.

In uniform circular motion, the forward force by the engine and the total resistive force against the motion of the car cancel each other out. These forces are not included in the diagram.

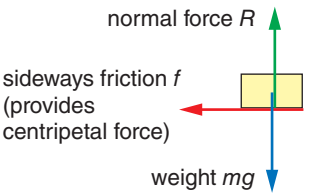


Fig 9.3b Forces acting on the car.

By $F = \frac{mv^2}{r}$, for constant v, $r\downarrow \Rightarrow F\uparrow$

If the car is going to turn a sharper corner (i.e. smaller r) at the same linear speed v, a larger centripetal force will be required. This demands a larger sideways friction (Fig 9.3c).

Simulation

→ This simulation simulates a car turning on a road under different conditions. It is available on OUP web in PC version (.exe) only.

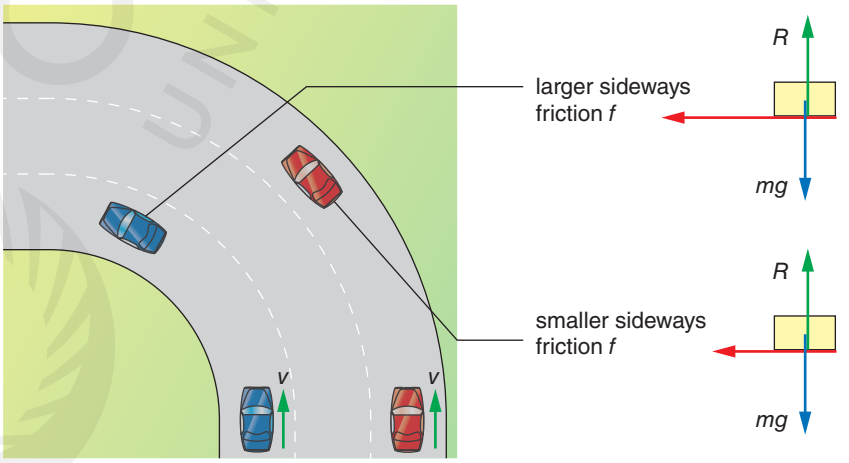


Fig 9.3c A larger sideways friction is required when a car turns a sharper corner.

By $F = \frac{mv^2}{r}$, for constant r, $v\uparrow \Rightarrow F\uparrow$

However, the friction between the car and the ground has a maximum value f_{max}. If the centripetal force required exceeds f_{max}, the car can no longer keep its circular motion but skids outwards (moves away from the centre of the circular path). This happens when a car turns a corner which is too sharp (Fig 9.3d on p.388) or when its speed is too high (Fig 9.3e on p.388).

Ext

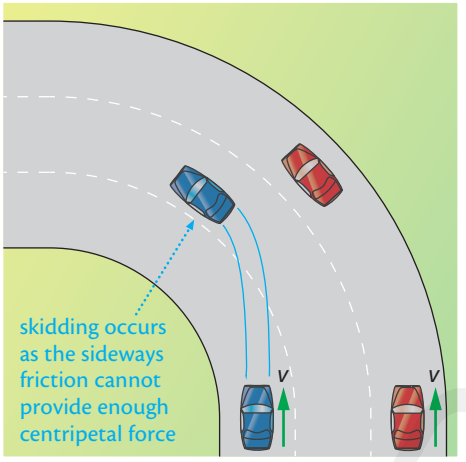


Fig 9.3d *Car X skids outwards because the centripetal force required exceeds f_{max}.*

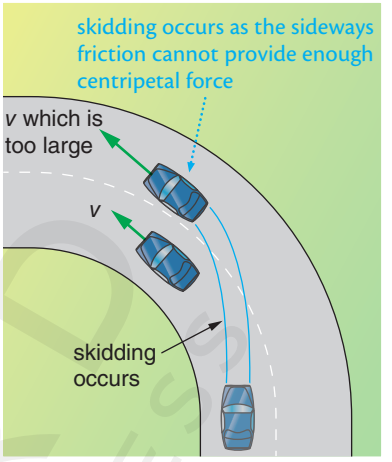


Fig 9.3e *A car will skid outwards at a corner if its speed is too high.*

Know more

Maximum friction between surfaces

The maximum friction f_{max} between two surfaces depends on the nature (e.g. dryness and roughness) of the surfaces and the normal force R between them. f_{max} and R is related by

$$f_{max} = \mu R$$

where μ is called the coefficient of friction and depends on the nature of the surfaces.

The value of μ is small when the surfaces are wet or smooth. That is why a car may skid more easily when it makes a turn on a wet road.

Example 3 Maximum speed for dry and wet roads

A car of 800 kg turns a corner of radius 20 m. The maximum sideways friction between the car and the road is 5500 N when the road is dry.

(a) Find the maximum speed v_{max} of the car without skidding.

(b) Why does the chance of skidding increase on rainy days?

Solution

(a) By $F = \dfrac{mv^2}{r}$,

$$v_{max} = \sqrt{\frac{F_{max} r}{m}} = \sqrt{\frac{5500 \times 20}{800}} = 11.7 \text{ m s}^{-1}$$

(b) The maximum friction between the car and the road decreases on rainy days. It is more likely that the friction is not large enough to provide the centripetal force required.

Tip

The car's speed attains v_{max} when F is maximum.

Ext

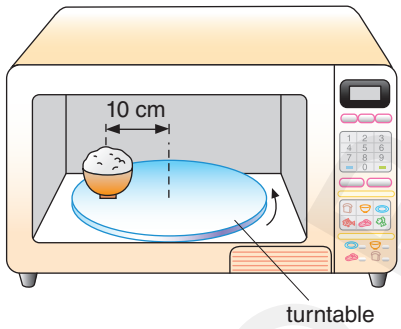
☆ HKDSE 2017 Paper 1B Q5

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Motion on a turntable

A bowl of rice of mass 0.8 kg is put in a microwave oven. It is 10 cm from the centre of the turntable. When the oven is turned on, the turntable rotates at a constant rate of 0.25 revolutions per second. The bowl of rice moves with the turntable without slipping.



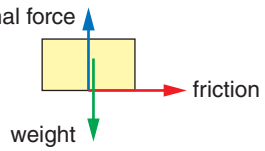
(a) Draw a free-body diagram for the bowl of rice at the instant shown. (2 marks)

(b) Find the acceleration of the bowl of rice. (3 marks)

(c) When the heating process ends, the turntable stops. The turntable is wet and is quite slippery. The bowl of rice slips for a distance of 2 cm after the turntable has stopped. Assume that the friction f acting on the bowl of rice is constant when it slips. Find f. (3 marks)

Solution

(a)



1A+1A

Common mistake
Students may wrongly write 'centripetal force' instead of 'friction'.

(b) $\omega = 0.25 \times 2\pi = 0.5\pi \text{ s}^{-1}$ 1M

Common mistake
Students may fail to apply the relationship 1 revolution = 2π.

Acceleration $= r\omega^2$

$= 0.1 \times (0.5\pi)^2$ 1M

$= 0.247 \text{ m s}^{-2}$ (towards the centre of turntable) 1A

(c) When the bowl of rice starts to slip,

$v = r\omega = 0.1 \times 0.5\pi = 0.05\pi \text{ m s}^{-1}$ 1M

Consider the motion from the moment that the bowl of rice starts to slip till it stops.

Loss in KE = work done against friction

$\frac{1}{2}m(u^2 - v^2) = fs$

Common mistake
Students may wrongly use the equation for circular motion to solve this part.

$\frac{1}{2} \times 0.8 \times [(0.05\pi)^2 - 0^2] = f \times 0.02$ 1M

$f = 0.493 \text{ N}$ 1A

▶ Revision 9 Q28 (p.411)

Ext

Writing practice

Question

A car makes a turn in a circular path. Explain why the chance of skidding increases when the car moves at a higher speed. (2 marks)

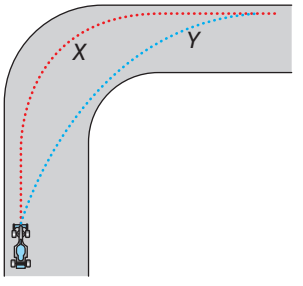
Answer

By $F = \frac{mv^2}{r}$, the car requires a larger centripetal force F when the speed v is higher. 1A

It is more likely that the friction is not large enough to provide the centripetal force required. 1A

Practise yourself

1 A race car is going to make a turn with a constant speed in car racing. If the driver wants to reduce the chance of skidding without lowering the car speed, which path, X or Y, should he take? Explain. (3 marks)



2 Making turns in the air

A model aeroplane flies straight ahead when its wings remain horizontal. However, it makes a turn when it banks (Fig 9.3f). Do you know why?

Video

→ This video shows how a tilted model aeroplane moves.

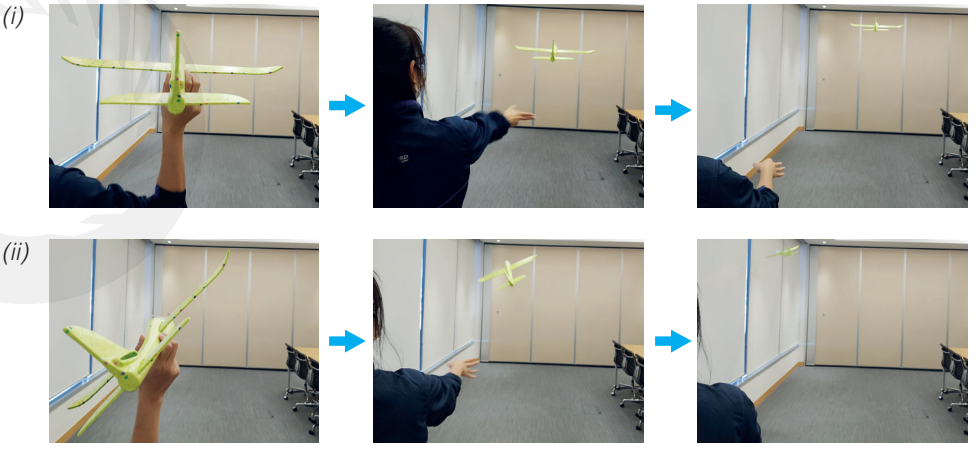


Fig 9.3f *(i) An aeroplane flies straight ahead when its wings are horizontal; (ii) it makes a turn when banking.*

Ext

When an aeroplane files, it is acted on by a lifting force L normal to its wings. If the wings are horizontal, L acts vertically upwards (Fig 9.3g). However, when the aeroplane banks, the horizontal component of L provides the centripetal force for the aeroplane to make a turn.

Suppose an aeroplane of mass m flies with a tilting angle θ and a constant speed v along a horizontal circular path of radius r (Fig 9.3h).

Thrust and air resistance cancel each other out and are not included in these diagrams.

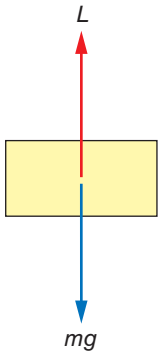


Fig 9.3g *Forces acting on a flying aeroplane when the wings are horizontal.*

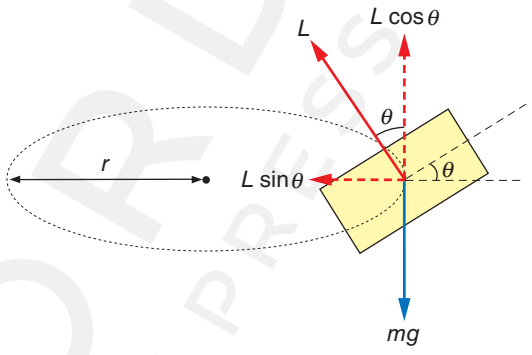


Fig 9.3h *Forces acting on an aeroplane when it is making a turn.*

In the **horizontal direction**, it is the horizontal component of the lifting force that provides the centripetal force:

$$L \sin\theta = \frac{mv^2}{r} \quad \ldots\ldots\ldots\ldots\ldots (1)$$

In the **vertical direction**, the vertical component of the lifting force and the weight balance each other:

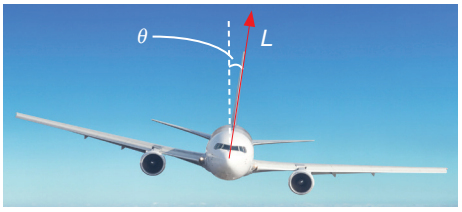
$$L \cos\theta = mg \quad \ldots\ldots\ldots\ldots\ldots (2)$$

Divide (1) by (2). The tilting angle θ is given by

$$\tan\theta = \frac{v^2}{gr}$$

Example 4 The turning of an aeroplane

An aeroplane of mass 4×10^6 kg makes a turn horizontally at a constant speed. The lifting force acting on it makes an angle θ to the vertical.



(a) Find the magnitude L of the lifting force if $\theta = 5°$.

(b) Find the centripetal force required to make the turn.

Ext

Solution

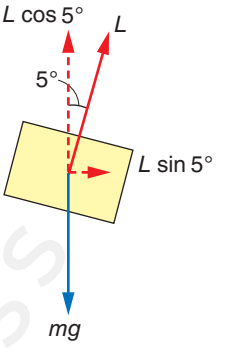
(a) Consider the vertical direction.

$L \cos \theta = mg$

$$L = \frac{mg}{\cos \theta}$$

$$= \frac{(4 \times 106) \times 9.81}{\cos 5°}$$

$$= 3.94 \times 10^7 \text{ N}$$



(b) Consider the horizontal direction.

Centripetal force

$= L \sin 5° = (3.94 \times 10^7) \times \sin 5° = 3.43 \times 10^6$ N

Tip

The lifting force provides the centripetal force.

Think about...

Now, the mass of the aeroplane is reduced. If it follows the same path to make a turn with the same speed, how should θ be changed?

3 Making turns on banked roads

The train in Figure 9.3i tilts when making a turn! Don't be surprised. The track is built at an angle and actually this can help the train make the turn more safely. This design is not only applicable to trains. Some roads are also built at an angle (Fig 9.3j).

Fig 9.3i A train is tilted when making a turn.

Fig 9.3j A banked road for turning.

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Ext

As discussed, the sideways friction provides the centripetal force for a car to make a turn on a level road. However, when the friction is not large enough, the car may skid outwards. The normal force acting on a car by a tilted road can contribute to the centripetal force. This reduces the reliance on friction between the road and a car (Fig 9.3k).

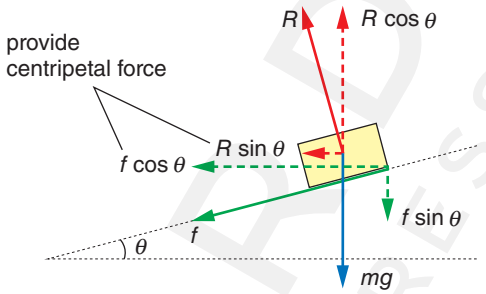


Fig 9.3k *The horizontal component of the normal force, as well as the sideways friction, provides the centripetal force.*

a Ideal banking

Consider a car turning a corner on a banked road along a circular path (Fig 9.3l). For an **ideally banked road**, the **horizontal component of the normal force provides the required centripetal force**. The car does not tend to move up or down the slope and there is **no sideways friction** acting on the car.

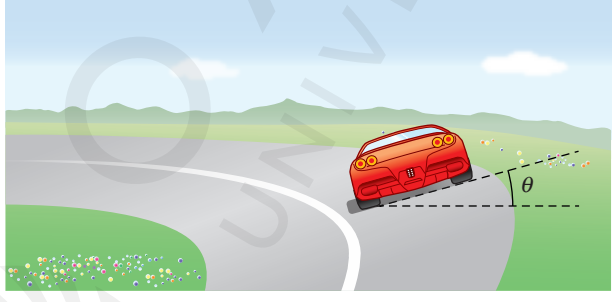


Fig 9.3l *A car making a turn on an ideally banked road.*

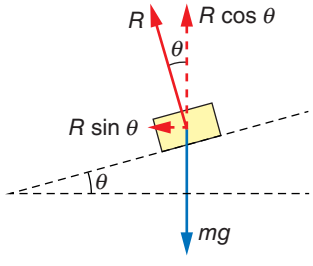


Fig 9.3m *Forces acting on the car.*

Suppose a car of mass m moves with a constant speed v on the road with an ideal banking angle θ along a circular path of radius r. In the **horizontal direction**, only the horizontal component of the normal force R provides the centripetal force (Fig 9.3m):

$$R \sin \theta = \frac{mv^2}{r} \quad \text{..... (3)}$$

In the **vertical direction**, the vertical component of R and the weight balance each other:

$$R \cos \theta = mg \quad \text{..... (4)}$$

Ext

Divide (3) by (4); the ideal banking angle θ is given by

$$\tan\theta = \frac{v^2}{gr}$$

We can see that θ is independent of the mass of the car. In practice, the banking angle of a road with a given r is determined based on the mean speed of the car passing along the road.

Example 5 | Ideal banking angle

A corner of a highway has a radius of curvature of 580 m. The mean speed of the cars passing along the highway is 20 m s^{-1}.

(a) Find the ideal banking angle of this corner.

(b) A car of mass 1500 kg turns the corner at 20 m s^{-1}. Find the normal force acting on it by the highway if the corner is banked at the ideal banking angle.

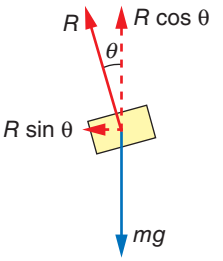
Solution

(a) Consider the horizontal direction.

$$R\sin\theta = \frac{mv^2}{r} \quad \text{..... (1)}$$

Consider the vertical direction.

$$R\cos\theta = mg \quad \text{..... (2)}$$



Divide (1) by (2):

$$\tan\theta = \frac{v^2}{gr}$$

$$= \frac{20^2}{9.81 \times 580}$$

$$\theta = 4.02°$$

The ideal banking angle is 4.02°.

(b) From (2):

$$R = \frac{mg}{\cos\theta}$$

$$= \frac{1500 \times 9.81}{\cos 4.02°}$$

$$= 14\,800 \text{ N}$$

The normal force acting on it by the highway is 14 800 N.

Ext

A similar calculation gives the same ideal banking angle for the case of a roller coaster car on a bend (Fig 9.3n) and *track cycling* (Fig 9.3o).

Fig 9.3n *A roller coaster car travelling around a bend.*

Fig 9.3o *Track cycling.*

b Non-ideal case

See p.394 for the meaning of 'mean speed'.

When a car turns on a banked road at a speed **higher than the mean speed**, the **centripetal force required becomes larger**. The horizontal component of the normal force alone is no longer sufficient to provide the centripetal force required. The car **tends to skid outwards**, so it is acted on by a **sideways friction *f* pointing inwards** (Fig 9.3p).

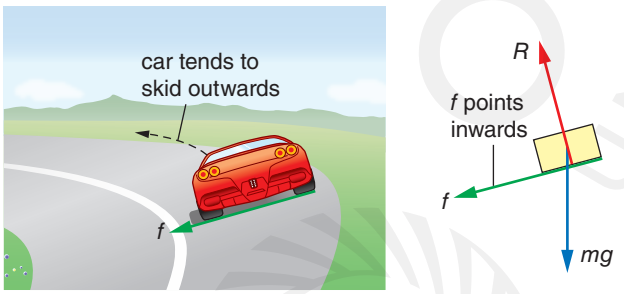


Fig 9.3p *As a car tends to skid outwards, a sideways friction f pointing inwards acts on the car to maintain its circular motion.*

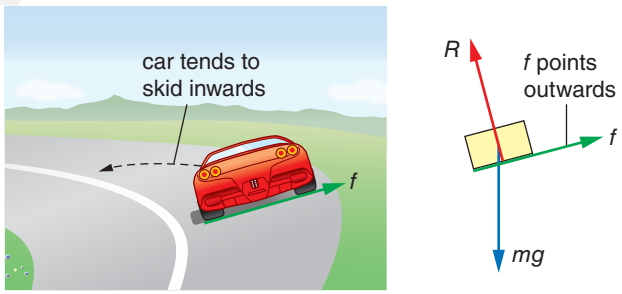


Fig 9.3q *As a car tends to skid inwards, a sideways friction f pointing outwards acts on the car to maintain its circular motion.*

Conversely, if a car makes the turn at a speed lower than the mean speed, the horizontal component of the normal force acting on the car is larger than the centripetal force required. The car tends to slide inwards for making a sharper turn. As a result, a **sideways friction *f* pointing outwards** will act on the car (Fig 9.3q).

In calculating the centripetal force for the car, both the normal force and the friction have to be considered.

track cycling 場地單車賽

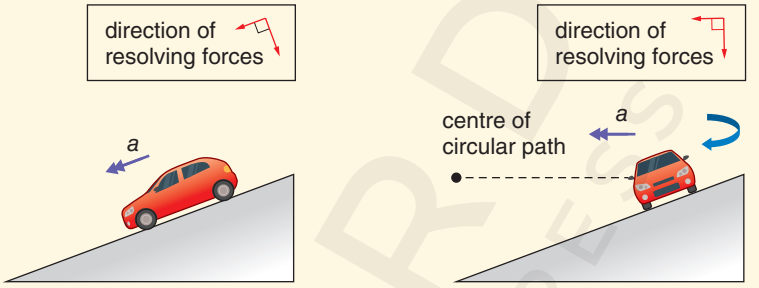
Ext

→ Teaching notes and exercises are available on OUP web.

Skill
Problem-solving

Resolving forces in different directions

When we solve problems involving an object sliding down a slope, we resolve forces into components parallel and perpendicular to the slope. However, in this chapter, we resolve forces into horizontal and vertical components. Why?



This is because the direction of acceleration of the object matters. In most cases in this chapter, the direction of acceleration is horizontal. Resolving forces into horizontal and vertical components can help simplify the calculations. You may refer to p.180 for the discussion about the direction of resolving forces and the direction of acceleration.

4 Rotor

The 'rotor' can be found in some amusement parks. It is basically vertical cylindrical barrels that can rotate at a high speed. Riders should stand against the wall before the rotor starts to rotate. When the rotor rotates fast enough, the floor is lowered but the riders remain 'stuck' to the wall (Fig 9.3r)!

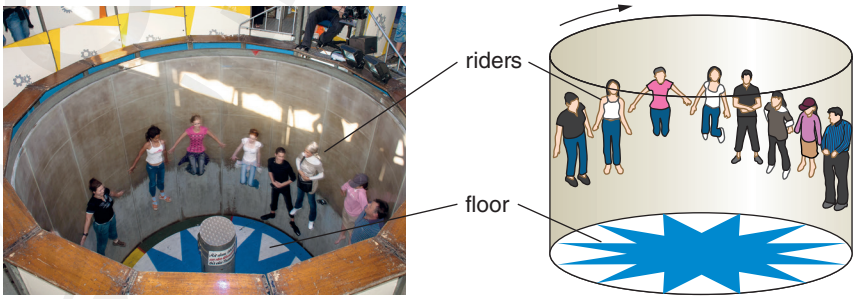


Fig 9.3r *Riders stuck to the wall of the rotor which is rotating at its full speed.*

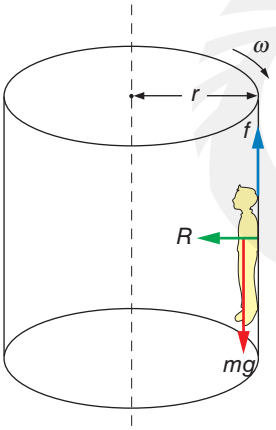


Fig 9.3s *Forces acting on a rider.*

Consider a rider of mass m inside a rotor of radius r rotating at a constant angular speed ω. When the floor is lowered, the rider is acted on by three forces, its weight mg, the normal force R and the friction f from the wall (Fig 9.3s).

In the **horizontal direction**, the normal force provides the centripetal force:

$$R = mr\omega^2$$

In the **vertical direction**, the forces balance each other:

$$f = mg$$

Ext

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

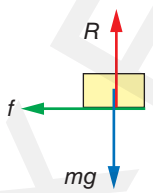
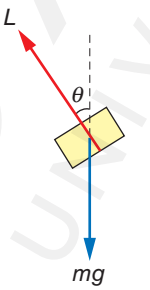
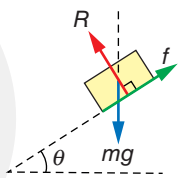
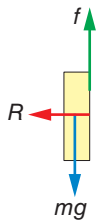
(For Q1–2). An object performs uniform circular motion **on a horizontal plane**.

1 Fill words in the blanks below to complete the descriptions about the object.

(a) The net force acting on the object along the horizontal direction is equal to _____.

(b) The net force acting on the object along the vertical direction is equal to _____.

2 The following shows the free-body diagrams of some objects moving in **horizontal circular paths** at constant speeds. The centripetal accelerations for the objects all point to the left at the moments shown. Complete the table with the symbols provided in the diagrams.

	Free-body diagram of object	Centripetal force	Force that balances the weight
e.g.		f	R
(a)			
(b)			
(c)			

Ext

Learn by practice 3

1 A car with a mass of 3000 kg is turning a corner of radius 100 m on a level road. If the maximum sideways friction on the car by the road is 40 000 N, what is the maximum speed that the car can turn the corner without skidding?

2 An airplane flies at a constant speed of 120 m s^{-1} along a horizontal circular path. The lifting force L acting on the plane is 180 000 N and it makes an angle of 20° to the vertical (Fig a).

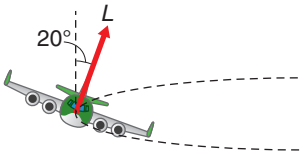


Fig a

(a) Find the mass of the plane.

(b) Find the radius of the circular path.

3 A 900-kg car moves around a corner with a radius of curvature of 20 m on a banked road (Fig b). The banking angle of the road is 30°. There is no friction acting on the car along the slope.

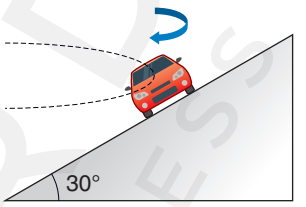


Fig b

(a) Find the normal force acting on the car by the road.

(b) Find the centripetal force acting on the car.

(c) Find the linear speed of the car during turning.

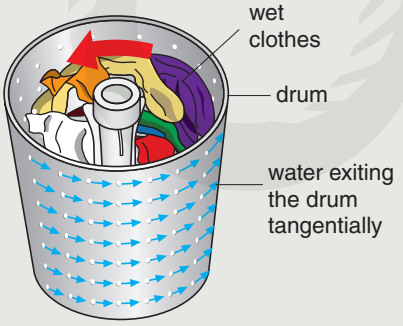
4 A corner on a highway has a radius of curvature of 200 m. The mean speed of the cars passing around this corner is 20 m s^{-1}. Find the ideal banking angle of this corner.

Know more

Devices for drying

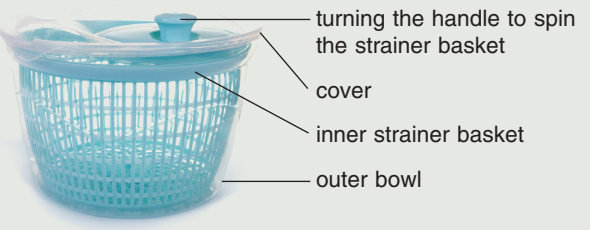
Drying clothes with a washing machine

To dry clothes, the drum of washing machine spins at a high speed. The normal forces from the wall of the drum provide the centripetal forces to keep the clothes rotating in the drum. On the other hand, water experiences no normal force when it reaches the holes on the wall, and is allowed to exit the drum through these holes.



Drying salad leaves with a salad spinner

A salad spinner is a kitchen tool for drying salad leaves. It works in a similar way as a washing machine. It usually consists of a bowl inside which there is a swivel strainer basket for placing washed salad vegetables. When the strainer basket spins, water on the salad is driven through the slits of the basket into the outer bowl.



Ext

5 Conical pendulum

Have you ever tried the swing ride in an amusement park (Fig 9.3t)? The riders are hung to swing along a circular path. Their motion is similar to that of a *conical pendulum* (Fig 9.3u).

The string of a conical pendulum never stays horizontal. Do you know why?

Fig 9.3t *Swing ride in an amusement park.*

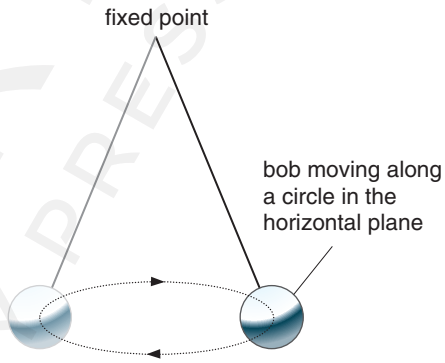


Fig 9.3u *A conical pendulum.*

Simulation

→ This simulation allows Ss to control the angular speed of a conical pendulum and find out how the angle of the conical pendulum is affected. It is available on OUP web in PC version (.exe) only.

Figure 9.3v shows a conical pendulum with a bob of mass m. The bob moves in a horizontal circular path of radius r at a constant angular speed ω. The string makes an angle θ with the vertical. The bob is acted on by two forces: its weight mg and the tension T in the string (Fig 9.3w).

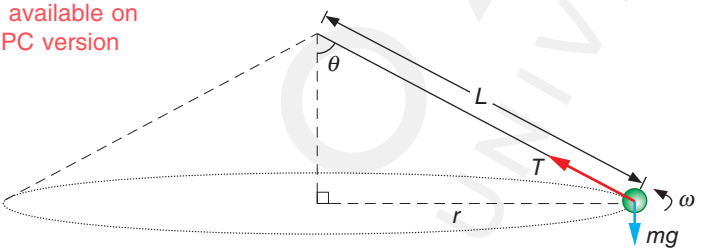


Fig 9.3v *The bob in a conical pendulum performs uniform circular motion.*

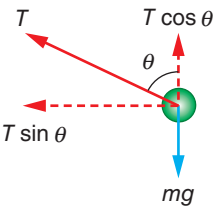


Fig 9.3w *Forces acting on the bob.*

In the **horizontal direction**, the horizontal component of T provides the centripetal force:

$$T \sin\theta = mr\omega^2 \quad \ldots\ldots\ldots\ldots\ldots (5)$$

In the vertical direction, the vertical component of T and the weight balance each other:

$$T \cos\theta = mg \quad \ldots\ldots\ldots\ldots\ldots (6)$$

mg is non-zero $\Rightarrow \cos\theta \neq 0 \Rightarrow \theta \neq 90°$

This shows why the string of a conical pendulum never stays horizontal. In fact, T must have a vertical component to balance mg so that the bob can move along a path that is horizontal.

Dividing (5) by (6), we have

$$\tan\theta = \frac{r\omega^2}{g}$$

conical pendulum 錐擺

Ext

Moreover, considering the triangle in Figure 9.3x, we have $\sin\theta = \frac{r}{L}$. Therefore, (5) becomes

$$T \times \frac{r}{L} = mr\omega^2$$

$$T = m\omega^2 L$$

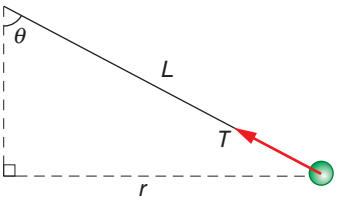


Fig 9.3x *Triangle for determining the trigonometric ratios.*

This equation can be verified by performing the following experiment. Both ω and L can be easily measured. The value of T can be fixed by making use of specially designed apparatus.

→ This video demonstrates Expt 9a.

Video

Experiment 9a / Verifying $T = m\omega^2 L$

1 Construct the centripetal force apparatus (Fig a). Measure the mass m of the rubber bung and the mass M of the screw nuts. Measure a length L of the nylon string from the rubber bung to the glass tube. Mark this length by sticking a paper marker on the string at the lower end of the glass tube.

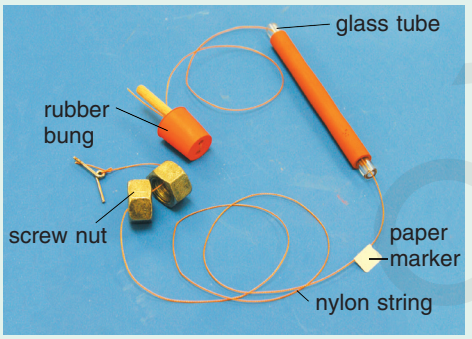
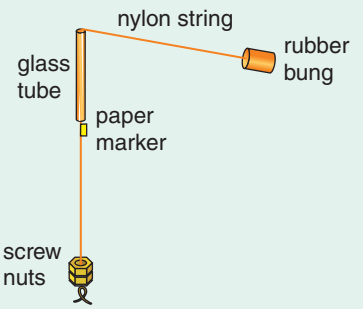


Fig a

Fig b



- Clear the area before the experiment.
- Wear safety goggles.

2 Whirl the rubber bung. The glass tube should be kept approximately vertical and at the same position. The paper marker should be kept at a position just below the glass tube (Fig b).

3 Measure the time taken for several revolutions and calculate the angular speed ω.

◀ Measuring the time for several revolutions instead of 1 can increase the accuracy of the value of ω.

4 Repeat steps 2 and 3 with different lengths L.

Discussion

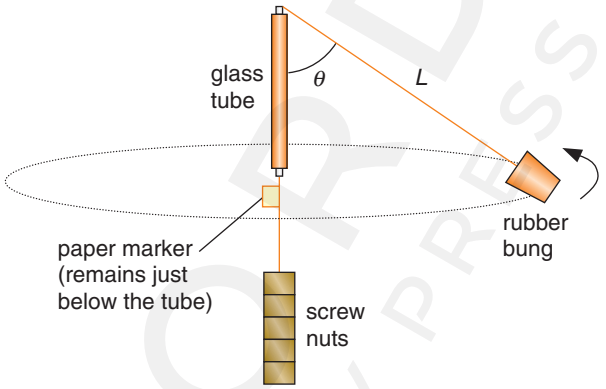
1 What is the tension T in the string as the rubber bung is whirling around?

2 Does $T = m\omega^2 L$ hold for all L?

Ext

Example 6 Verifying $T = m\omega^2 L$ with the centripetal force apparatus

Refer to the set-up in Experiment 9a. The mass m of the rubber bung is 20 g. The total mass M of the screw nuts is 1 kg. The length L of the string is at an angle θ to the vertical.



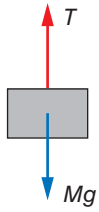
(a) Find the tension T in the string.

(b) How would T and θ change if the experiment is repeated with a smaller L? Explain briefly.

(c) Show that $T = m\omega^2 L$. Hence, determine how ω would change if the experiment is repeated with a smaller L.

Solution

(a) As the screw nuts remain at rest, the net force acting on them is zero.



$T = Mg$

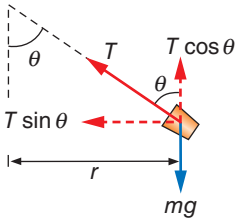
$= 1 \times 9.81$

$= 9.81$ N

Tip

It is important to identify the quantities which have fixed values in a certain situation.

(b) T would remain unchanged as it is still equal to the weight of the screw nuts.



$T \cos\theta$ is still equal to the weight of the rubber bung. As T remains unchanged, θ also remains unchanged.

(c) Consider the horizontal direction.

$T \sin\theta = mr\omega^2$

$T \sin\theta = m \times (L \sin\theta) \times \omega^2$

$T = m\omega^2 L$

According to the equation, ω would become larger if L becomes smaller.

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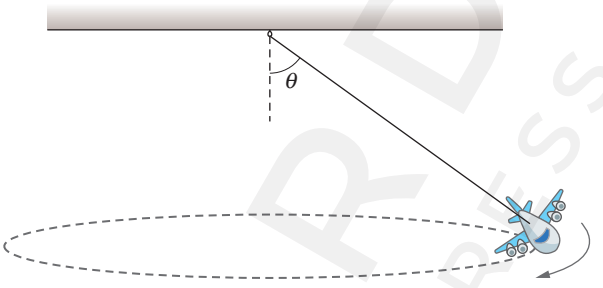
Ext

Example 7 Toy plane tied to the ceiling

A toy plane of mass m is whirled in a horizontal circular path of radius r at a constant angular speed ω. The tension in the string is T and the string makes an angle θ with the vertical.



(a) Show that $\cos\theta = \dfrac{g}{\omega^2 L}$.

(b) How will θ change if the toy plane is whirled more quickly?

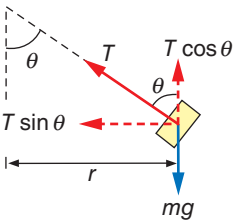
Solution

(a) Consider the horizontal direction.

$T\sin\theta = mr\omega^2$

$T\sin\theta = m \times (L\sin\theta) \times \omega^2$

$T = m\omega^2 L$ .....(1)



Consider the vertical direction.

$T\cos\theta = mg$ .....(2)

Substitute (1) into (2):

$m\omega^2 L \times \cos\theta = mg$

$$\cos\theta = \frac{g}{\omega^2 L}$$

Tip

According to $\cos\theta = \dfrac{g}{\omega^2 L}$,

for constant L,

$\omega \uparrow \Rightarrow \cos\theta \downarrow \Rightarrow \theta \uparrow$

(b) θ will increase.

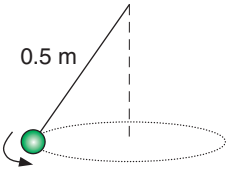
Learn by practice 4

1 A ball of 0.3 kg is attached to a string with a length of 0.5 m. The other end of the string is fixed to the ceiling. The ball moves at a constant speed along a horizontal circular path. The string makes an angle of 30° with the vertical.

(a) Find the tension in the string.

(b) Find the radius of the path.

(c) Find the angular speed of the ball.



Ext

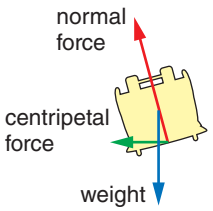
Exercise 9.3

★**1** A roller coaster car turns on a banked rail (Fig a). The centripetal force it needs is equal to the horizontal component of the normal force from the rail.

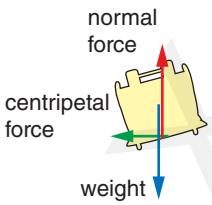
Fig a

Which of the following is the free-body diagram for the car when it is viewed from the front?

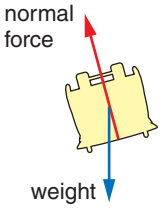
A



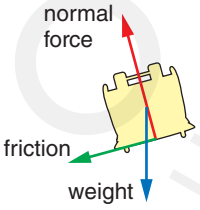
B



C



D



★**2** The maximum safe speed of a car driving around a corner on a dry level road is 30 m s^{-1}. The maximum sideways friction between the road and the car is halved when the road is wet. What is the maximum safe speed for the car to drive around the corner when the road is wet?

A 7.07 m s^{-1} B 14.1 m s^{-1}

C 15.0 m s^{-1} D 21.2 m s^{-1}

★**3** A bus of mass 15 000 kg attempts to turn on a level road at a speed of 10 m s^{-1}. The radius of curvature of the turn is 50 m. If the maximum sideways friction between the bus and the road is 60 000 N, will the bus skid when turning? Show your calculation.

★**4** Two dishes of dim sum, *X* and *Y*, each of mass 0.8 kg, are placed on a horizontal lazy Susan (Fig b). *X* and *Y* are respectively 20 cm and 40 cm away from the centre of rotation. The maximum friction between each dish and the lazy Susan is the same. When the lazy Susan rotates constantly with a period of 8 s, *X* and *Y* move with it without slipping.

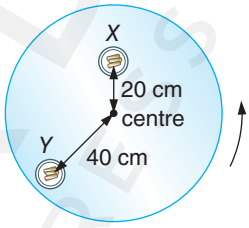


Fig b

(a) Find the friction acting on *X* and *Y*.

(b) The lazy Susan rotates faster and faster. Which dish will slip first? Explain briefly.

★**5** A 800-kg car makes a turn on a level road with a speed of 25 m s^{-1}. The radius of curvature of the turn is 70 m.

(a) Find the centripetal force acting on the car.

(b) The car then tries to take a shortcut by making the turn in a circular path of smaller radius *r* without changing the speed. Given that the maximum friction between the road and the car is 8730 N, find the minimum value of *r*.

★**6** A 0.8-kg ball tied to a string is whirled around at a constant speed in a horizontal circle of radius 0.5 m. The string makes an angle of 20° to the vertical (Fig c).

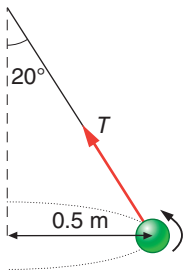


Fig c

(a) Find the tension *T* in the string.

(b) Find the centripetal force for the ball.

(c) Find the linear speed of the ball.

★**7** An aircraft makes a turn along a horizontal circular path at a constant speed of 200 m s^{-1}. The radius of the circular path is 10 km.

(a) Where does the centripetal force that a passenger in the aircraft needs come from?

(b) Find the ratio of the centripetal force that the aircraft needs to the aircraft's weight.

(c) Find the angle between the aircraft's wings and the horizontal.

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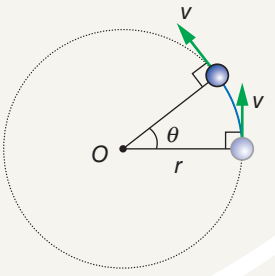
Ext

Key concepts 9

Describing uniform circular motion



Angular displacement θ

- unit: rad
- $2\pi = 360°$

Angular velocity ω

- $\omega = \dfrac{\theta}{t}$
- unit: rad s^{-1}
- $v = r\omega$

Centripetal force F

- provided by the net force acting on the object
- pointing towards centre
- $F = \dfrac{mv^2}{r} = mr\omega^2$

Centripetal acceleration a

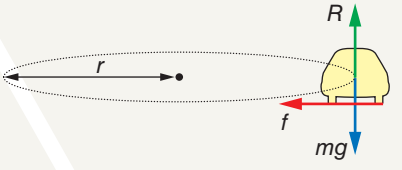
- pointing towards centre
- $a = \dfrac{v^2}{r} = r\omega^2$

Period $T = \dfrac{2\pi r}{v} = \dfrac{2\pi}{\omega}$

Examples of uniform circular motion

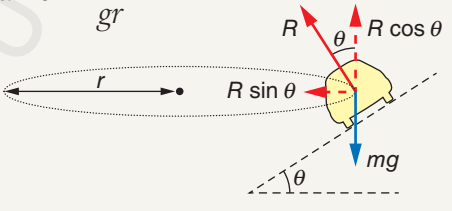
1 Making turns on level roads

- centripetal force = $f = \dfrac{mv^2}{r}$
- $R = mg$



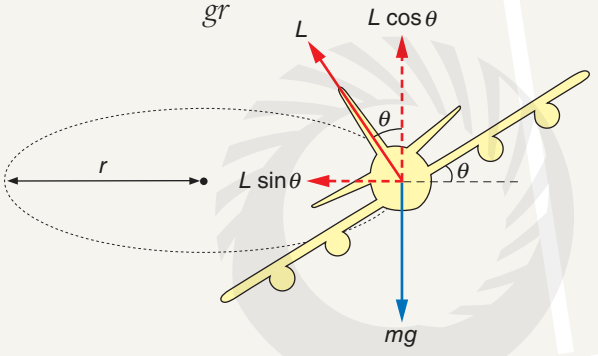
2 Making turns on banked roads (ideal banking angle)

- centripetal force
  $= R \sin\theta = \dfrac{mv^2}{r}$
- $R \cos\theta = mg$
- $\tan\theta = \dfrac{v^2}{gr}$



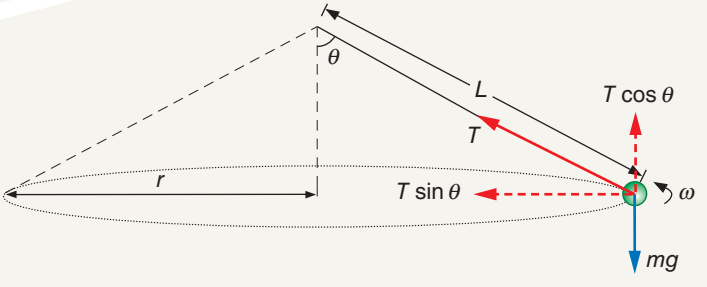
3 Making turns in the air

- centripetal force
  $= L \sin\theta = \dfrac{mv^2}{r}$
- $L \cos\theta = mg$
- $\tan\theta = \dfrac{v^2}{gr}$



4 Conical pendulum

- centripetal force = $T \sin\theta$
- $T \cos\theta = mg$



Ext

Key terms

1 angular displacement 角位移 p.371
2 angular speed 角速率 p.373
3 angular velocity 角速度 p.373
4 centripetal acceleration 向心加速度 p.379
5 centripetal force 向心力 p.378
6 period 週期 p.375
7 radian (rad) 弧度 p.370
8 uniform circular motion 勻速圓周運動 p.373

Revision 9

Multiple-choice questions

★1 Misc The magnetic disk of a hard disk drive spins at 5400 rpm (rotations per minute). The diameter of the magnetic disk is 6.35 cm.

Fig a

What is the magnitude of the acceleration of a point on the outside edge of the disk?

A 10 200 m s^{-2} B 20 300 m s^{-2}
C 25 700 m s^{-2} D 51 400 m s^{-2}

★2 9.2 A dish of dim sum rotates on a lazy Susan at a constant angular speed ω (Fig b). Which arrows correctly show the directions of the linear velocity v and the acceleration a of the dim sum?

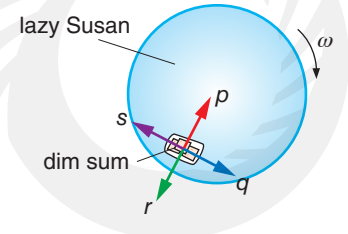


Fig b

	v	a
A	q	r
B	q	p
C	s	p
D	s	s

★3 Misc X and Y are two points on a fidget spinner (Fig c).

Fig c

When the spinner rotates about point O, X and Y must have the same

(1) angular speed.
(2) linear speed.
(3) centripetal acceleration.

A (1) only B (1) and (2) only
C (2) and (3) only D (1), (2) and (3)

★4 Misc Music will be played when a record, which is a flat disc, is rotated by a player (Fig d). Suppose there is an ant on a record which rotates at a constant angular speed. Which of the following will decrease when the ant walks towards the centre of the disc?

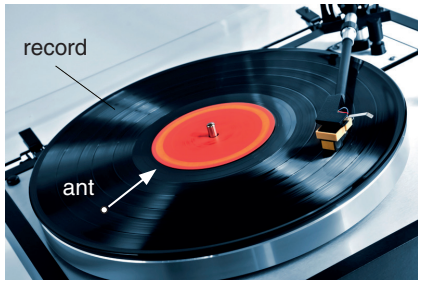


Fig d

(1) The ant's angular speed
(2) The ant's linear speed
(3) The net force acting on the ant

A (1) only B (2) only
C (1) and (3) only D (2) and (3) only

Ext

★5 Misc

X, Y and Z are three athletes running steadily on two circular tracks in an anticlockwise direction (Fig e). The positions of X and Y are always aligned with the centre O of the circular tracks. Y and Z run at the same linear speed.

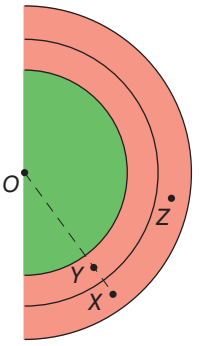


Fig e

Which of the following statements are correct?

(1) X will overtake Z.

(2) Z has a larger acceleration than X.

(3) The accelerations of X and Y always point in the same direction.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★6 9.3

A doll is hung by a string from the rear-view mirror in a car. When the car turns a corner at a constant speed on a level road, the doll swings to the left and the string hanging the doll makes an angle θ to the vertical as viewed from the back seat (Fig f). In which direction is the car turning? What is the centripetal acceleration a of the car?

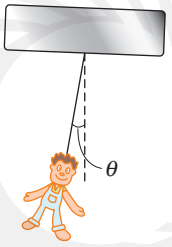


Fig f

	Turning direction	a
A	right	$g \tan \theta$
B	right	$g \sin \theta \cos \theta$
C	left	$g \tan \theta$
D	left	$g \sin \theta \cos \theta$

★★7 9.3

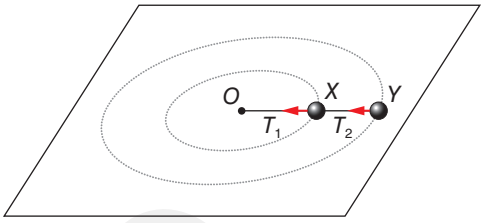


Fig g

Two identical bodies, X and Y, are connected with strings to a point O on a smooth horizontal plane (Fig g). They revolve steadily at the same angular speed about O. The tension in the string connecting O and X is T_1 while the tension in the string connecting X and Y is T_2. The two strings have the same length. Find the ratio of T_1 to T_2.

A 1 : 2 B 2 : 3

C 1 : 1 D 3 : 2

Refer p.160, 397

9.3 **8 HKDSE 2015 Paper 1A Q10**

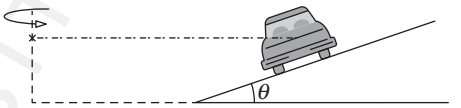


Fig h

Figure h shows the rear view of a car of mass m which travels along a circular road banked with an angle θ to the horizontal. The car moves at a certain speed such that it experiences no frictional force along the inclined surface. Which of the following represents the centripetal force on the car?

A $mg \sin \theta$ (31%) B $mg \sin \theta \cos \theta$ (21%)

C $\dfrac{mg \cos \theta}{\sin \theta}$ (15%) D $\dfrac{mg \sin \theta}{\cos \theta}$ (33%)

9.2 **9 HKDSE 2016 Paper 1A Q13**

Particles P and Q are fixed at distances r and $\frac{r}{2}$ respectively from the centre O of a horizontal circular platform which is rotating uniformly as shown.

The ratio of the acceleration of P to that of Q is

A 1 : 2.

B 2 : 1. (45%)

C 1 : 4.

D 4 : 1.

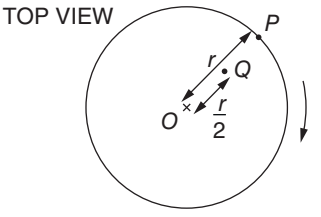


Fig i

Q8 Exam Report: About one-third of the candidates were able to find the centripetal force by resolving the components of the normal force.

Ext

9.2 **10** **HKDSE 2020 Paper 1A Q9**

Particles P and Q are fixed on the surface of a sphere rotating about a vertical axis passing through the centre O of the sphere as shown. The horizontal shaded region X divides the sphere into two halves. P is at the edge of region X while Q is at an angle of elevation θ above region X.

Find the ratio of the centripetal acceleration of P to that of Q.

A $1 : \cos\theta$ (59%)

B $1 : \sin\theta$

C $\cos\theta : 1$

D $\sin\theta : 1$

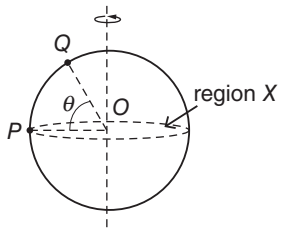


Fig j

Conventional questions

★**11** 9.1 A wind turbine (Fig k) rotates steadily at 15 revolutions per minute. Each blade is 20 m long.

Fig k

(a) Find the angular speed of the blades. (2 marks)

(b) Find the linear speed of the tip of a blade. (2 marks)

(c) How long does it take for a blade to rotate 60°? (2 marks)

★**12** Misc A person of mass 60 kg stands in Hong Kong which is located at latitude 22.3° North (Fig l). The person can be treated as a point mass.

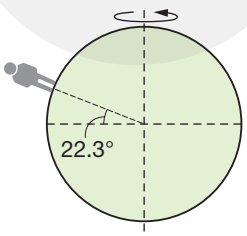


Fig l

It is known that the Earth rotates once in 24 hours and the radius of the Earth is 6.37×10^6 m.

(a) Estimate the angular speed of the person as the Earth rotates. (2 marks)

(b) Estimate the centripetal acceleration of the person. (2 marks)

(c) Estimate the centripetal force for the person. Draw an arrow to show the direction of this force in Figure l. (3 marks)

(d) Is there friction acting on the person by the ground? (1 mark)

★**13** 9.3 A man is driving along a straight road at speed v_0 and he suddenly finds himself coming to a T-junction. He is at a distance s away from a river (Fig m). With maximum braking, the car would stop with its nose just overhanging the riverbank. Assume the acceleration of the car is uniform during braking.

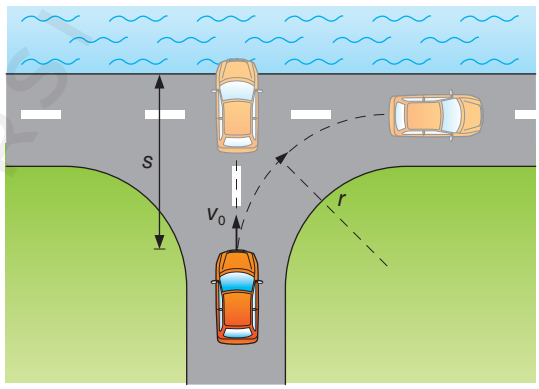


Fig m

(a) Find the acceleration of the car during braking. Express your answer in terms of v_0 and s. (2 marks)

(b) Suppose the car turns in a circular path of radius r at a constant speed v_0. Assume the maximum sideways friction providing the centripetal force for the car is the same as the maximum braking force.

(i) Find the minimum value of r. (2 marks)

(ii) With the above assumptions, is it safe for the driver to make a turn? (1 mark)

Ext

***14** 9.3 A skateboarder of mass 70 kg performs horizontal uniform circular motion on the vertical wall of a skate park of radius 10 m (Fig n). The maximum friction between him and the wall is $0.6R$, where R is the normal force acting on him by the wall.

Fig n

(a) Draw a free-body diagram for the skateboarder. (2 marks)

(b) Find the friction acting on the skateboarder. (2 marks)

(c) Find the minimum normal force acting on the skateboarder. Hence, find the minimum linear speed of the skateboarder. (3 marks)

***15** 9.2 An astronaut takes part in high-G training which aims to prevent him from losing consciousness when experiencing high levels of acceleration during a space flight. He sits at one end of the 20-G centrifuge (Fig o) which spins horizontally very quickly with a constant speed. He is 8.84 m away from the axis of rotation.

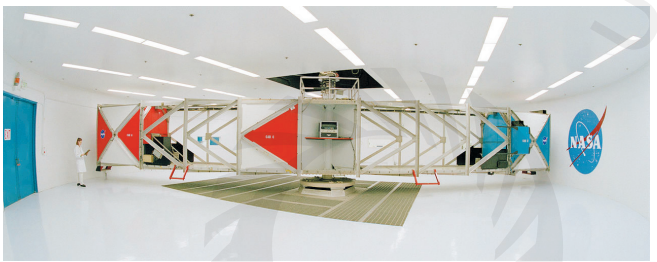


Fig o

(a) Find the angular speed of the astronaut when the net force acting on him is 20 times his weight. (2 marks)

(b) How would the magnitude of the net force acting on him change if his distance from the axis of rotation is reduced while the rate of rotation of the centrifuge remains unchanged? Explain briefly. (2 marks)

***16** 9.3 A roller coaster train makes a turn (Fig p). Assume that the turn is horizontal and its radius of curvature is 8 m. The train moves at a constant speed of 10 m s^{-1} when it makes the turn.

Fig p

John is a passenger on the train and his mass is 65 kg. He is acted on by two forces: his weight and the force F acting on him by the train.

(a) Find the centripetal force on John. (2 marks)

(b) Find the magnitude of F. (3 marks)

***17** 9.2 Peter, Paul and Mary stand at the equator, latitude 60° N and the North Pole respectively (Fig q). Assume that the Earth is a sphere of radius 6370 km, rotating about an axis passing through the North Pole and the South Pole. Assume that the Earth rotates once in 24 hours and each person can be treated as a point mass.

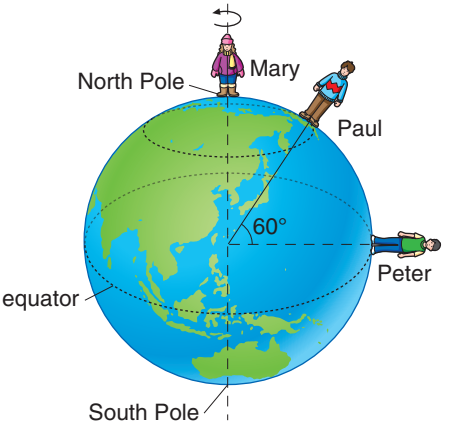


Fig q

(a) Find the angular speed and linear speed of each person. (4 marks)

(b) Find the magnitude of the centripetal acceleration of each person. (3 marks)

(c) Draw an arrow to show the direction of the centripetal acceleration of each person on Figure q. (2 marks)

Ext

***18** 9.3 Figure r shows a boy playing a tetherball. The ball of mass 1.5 kg is attached to a vertical pole by a light string of length 1.2 m. Consider a certain time period in which the ball moves at 4 rad s^{-1} along a horizontal circular path.

Fig r

(a) Find the tension T in the string. (3 marks)

(b) Find the angle θ between the string and the pole. (2 marks)

(c) How will T change if the ball moves at a higher angular speed? Explain your answer. (2 marks)

(d) How will θ change if the ball moves at a higher angular speed? (1 mark)

***19** 9.1 Cable cars can be found in San Francisco. When the cable car reaches the end of the line, it is turned around using a turntable (Fig s).

Fig s

Figure t shows the top view of the car during turning. Point O is the centre of the turntable and X is a point at the front end of the car. Assume that it takes 8 s for the car to turn 90° and X moves at a constant speed of 0.5 m s^{-1}.

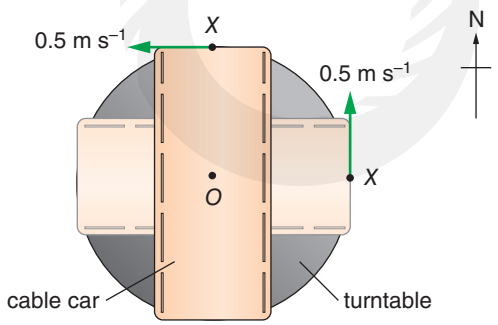


Fig t

(a) Find the angular speed of the car. (2 marks)

(b) Find the distance of X from O. (2 marks)

(c) A person walks from O towards X during turning. How do his angular speed and linear speed change? (2 marks)

(d) In Figure t, the velocities of X before and after the car has turned by 90° are shown. In which direction does the change in velocity point to? (1 mark)

***20** Misc Turn 8 at the Istanbul Park racing circuit is a famous corner in car racing (Fig u). Assume a car turns this corner in a horizontal circular path at a constant speed of 76 m s^{-1} and the driver experiences an acceleration of $5g$ for 7 s.

Fig u

(a) Find the radius of the corner. (2 marks)

(b) Find the magnitude of the car's angular displacement in turning the corner. (3 marks)

(c) The mass of the driver's head with his helmet is about 6 kg. Estimate the magnitude of the force acting on his head by his neck in turning the corner. (3 marks)

***21** 9.3 An object is hung by a string from point X in a car (Fig v). The string will become inclined at an angle θ to the vertical when the car turns around a corner.

Fig v

(a) Why does the string become inclined when the car turns around a corner? (2 marks)

(b) Suppose a car turns around a corner in a horizontal circular path at a constant speed of 15 m s^{-1}. The radius of the path is 50 m.

(i) Find θ. (3 marks)

(ii) How will θ change if the object is replaced by a more massive one? (1 mark)

Ext

***22** 9.3 When a rotor of radius 2 m rotates at an angular speed ω, a rider of mass 50 kg can just stick to the wall without falling (Fig w). The normal force R acting on the rider by the wall is related to the maximum friction f_{max} between them by $f_{max} = 0.5R$.

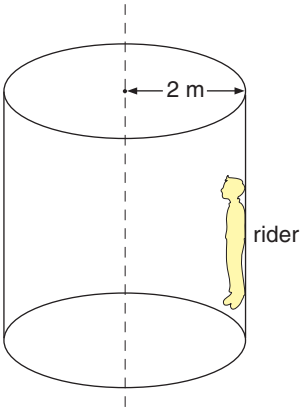


Fig w

(a) Find the friction f between the rider and the wall. (2 marks)

(b) (i) Find the centripetal force for the rider. (2 marks)

(ii) Find ω. (2 marks)

(c) There is another rider of a larger mass in the rotor. Can this rider remain stuck to the wall? (1 mark)

(d) Explain what would happen to the rider if the angular speed of the rotor decreases. (3 mark)

***23** 9.3 A ball of mass 50 g moves steadily at 1.2 m s^{-1} along a horizontal circular path on the smooth inner surface of a cone (Fig x).

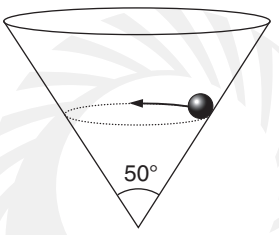


Fig x

(a) Draw all the forces acting on the ball in Figure x. (2 marks)

(b) Find the magnitudes of all the forces in (a). (3 marks)

(c) Find the radius of the circular path. (2 marks)

(d) Can the ball move in the same path at a higher speed? (1 mark)

***24** 9.3 Roy spins his son at a constant speed along a horizontal circular path (Fig y). Their arms make an angle θ to the vertical. Roy rotates along the vertical line passing through point A. F is the force acting on the boy by Roy.

Fig y

Consider the boy as a point mass located at his c.g. The mass of the boy is 30 kg. His c.g. is 1.5 m from A. The angle θ is 60°.

(a) Find the force F acting on the boy by Roy. (2 marks)

(b) Find the boy's angular speed. (3 marks)

(c) If Roy wants to spin his son at a higher angular speed, how should F and θ change? (2 marks)

****25** Misc A turntable can be rotated steadily at different speeds by an electric motor about the vertical axis as shown (Fig z). A trolley is connected to a point on the turntable by an inextensible horizontal string. The tension acting on the trolley by the string is measured by the force sensor in the trolley. The trolley's wheels are parallel to the radius of the turntable. A stop-watch is also available.

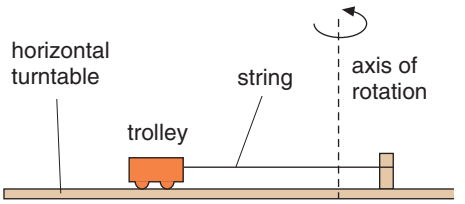


Fig z

Describe how you would use the above apparatus to examine the relationship between the centripetal force F that the trolley needs and its angular speed ω. Write down your expected result. (6 marks)

▶ Refer p.401

Ext

**26 9.3 Block B of mass 0.5 kg is connected by a light spring to the centre O of a horizontal turntable (Fig aa). When the turntable rotates about O at an angular speed ω, OB is 10 cm and the spring pulls B with a force T of 3 N. The maximum friction between B and the turntable is 2 N.

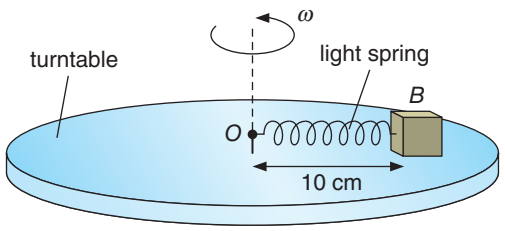


Fig aa

(a) Suppose the turntable rotates at the minimum angular speed ω_{min} so that OB remains at 10 cm.

(i) Find ω_{min}. (2 marks)

(ii) What would happen to the block if the spring suddenly broke? Explain briefly. (2 marks)

(b) Suppose the turntable rotates at the maximum angular speed ω_{max} so that OB remains at 10 cm.

(i) Find ω_{max}. (1 mark)

(ii) What would happen to the block if the spring suddenly broke? Explain briefly. (2 marks)

Refer p.395

**27 Misc A boy sits on the edge of a merry-go-round of radius 1.5 m (Fig ab).

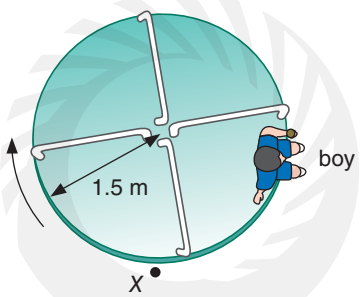


Fig ab

The merry-go-round rotates uniformly for a period of 2 s. The boy throws a stone vertically upwards from a height 1 m above the ground when he reaches point X. The maximum height the stone reaches is 3 m above the ground. The stone then falls onto the ground.

(a) Find the linear speed of the boy. (2 marks)

(b) Find the time of flight of the stone. (3 marks)

(c) Find the horizontal distance between the landing position of the stone and X. (2 marks)

Refer p.384

9.3 **28 HKDSE 2017 Paper 1B Q5**

The photo shows a turntable commonly used in restaurants.

Fig ac

A teapot of mass 1 kg is put 30 cm from the centre of a horizontal turntable. Figure ad shows the side view. When the turntable is rotating, the teapot remains at the same position on the turntable.

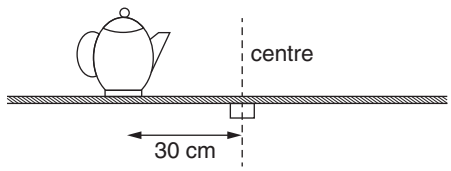


Fig ad

(a) On Figure ad, draw and label all the forces acting on the teapot when the turntable is rotating. (2 marks)

(b) Taking the teapot as a point mass, estimate the net force acting on the teapot when the turntable is rotating at a rate of 0.5 revolutions per second. (3 marks)

(c) The turntable is suddenly stopped and the teapot slips. The turntable is rotating at a rate of 0.5 revolutions per second just before it stops, and the frictional force acting on the teapot is 10 N when it is slipping. Determine the distance travelled by the teapot after the turntable stops. (3 marks)

Q28 Exam Report: This question tested candidates' knowledge and understanding on circular motion.
(a) Most candidates managed to indicate the forces acting on the teapot although a few of them labelled the frictional force as 'centripetal force'.
(b) Quite a lot of the candidates failed to work out the correct angular velocity from the rate of revolution given.
(c) Weaker ones misunderstood this part and they wrongly applied the equation for circular motion to tackle the problem.

Ext

Misc 29 HKDSE 2019 Paper 1B Q4

(a)

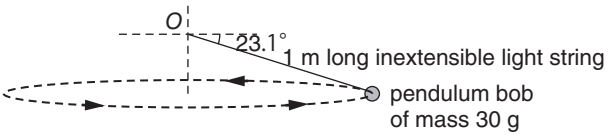


Fig ae

A pendulum bob of mass 30 g is tied to a fixed point O by a 1 m long inextensible light string. It is swirled to describe a horizontal circle uniformly at an angular velocity of 5.0 rad s^{-1} as shown in Figure ae. Neglect air resistance. (g = 9.81 m s^{-2})

(i) What is the bob's rotation rate (in revolutions per second)? (1 mark)

(ii) Indicate on Figure ae the centripetal force F_C required for the motion of the bob. Find F_C. (3 marks)

(iii) Explain whether the magnitude of the tension in the string is greater than, equal to or smaller than the centripetal force F_C found in (b). (2 marks)

(b) The moon is orbiting around the Earth uniformly in a circular path under the influence of the Earth's gravitational attraction.

(i) Explain why the speed of the moon remains unchanged although it is acted upon by gravitational force. (2 marks)

(ii) A student claimed that as the moon is much less massive than the Earth, it exerts negligible force on the Earth. Comment on the student's claim. (2 marks)

Q29 Exam Report: Candidates' performance was fair.

(a) (i) Quite a lot of them had difficulties relating the angular velocity with the rotation rate.

(ii) Most candidates were able to find the centripetal force F_C. However, a few did not indicate F_C in the figure or mistook the length of the string to be the radius.

(iii) A majority of the candidates tackled this part by finding out the value of the tension. Weaker candidates held that the tension was the resultant of the bob's weight and the centripetal force.

(b) (i) Some candidates failed to point out that no work is done by the gravitational force acting on the Moon while weaker ones failed to realize that the centripetal force is actually the gravitational pull and stated that the net force acting on the Moon was zero.

(ii) Most were able to mention an action-and-reaction pair or use F = to explain.

Know more

Glass trick

If you are given a glass, can you use it to lift a small ball off the table (Fig a)?

Here is the trick. Place the glass upside down over the ball (Fig b). Move the glass in circles quickly to spin the ball. If you spin fast enough, the ball will roll up the wall of the glass and perform circular motion. The normal force acting on the ball by the wall provides the centripetal force required (Fig c). Keep spinning and you can lift the glass off the table with the ball rolling inside.

Is it possible to perform this trick using the cup in Figure d? Why?

Fig a

Fig b

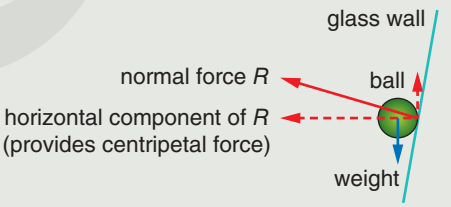


Fig c

Fig d